

IECEX/ATEX Certification Information Requirements

Gas Leak Detector (GLD V2)

Manufacturer: PT Galaksi Megatama Indonesia

Revision: 25 September 2026

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1. BASIC INFORMATION (APPLICATION AND ORGANIZATION)

1) Application form (provided by ExCB as a template)

Issued by the ExCB as a template; already received.

Status: Template received from the ExCB.

2) Manufacturer's business license/company registration certificate

The Manufacturer named in this application is PT Galaksi Megatama Indonesia, which holds the Indonesian industrial manufacturing business license (NIB) relevant to fabricating the equipment.

Particular	Value
Registered legal name	PT Galaksi Megatama Indonesia
Legal form	Perseroan Terbatas (limited liability company), National Private / PMDN, Small Enterprise scale
Business registration number (NIB)	0510220020593, issued 5 Oct 2022 (OSS system)
Taxpayer ID (NPWP)	61.126.496.1-427.000, registered 29 Sep 2022
Deed of establishment	Deed No. 52, 22 Sep 2022 (Notary Steffi Alphanie, S.H., M.Kn., Cilegon City)
Ministerial ratification	MOLHR Decree No. AHU-0067398.AH.01.01.Year 2022 (29 Sep 2022)
Registered (domicile) address	Plaza Summarecon Bekasi, Jl. Bulevar Ahmad Yani Kav. K.01, Level 7, Bekasi City, West Java 17143
Relevant KBLI (business scope)	25119, 25120 (tanks/reservoirs/containers of metal, incl. pressurized gas cylinders), 28221, 25920

3) Manufacturer's organizational chart and contact information

The statutory Board of Directors and Board of Commissioners are known from the deed of establishment:

Position	Name
President Director	Nur Rohman
Director	Antonius Prasetyo
Commissioner	Crisa Andi Sujatmiko

4) Address of manufacturing plant and brief introduction of production facilities

Registered office and NIB-registered production sites of the Manufacturer, PT Galaksi Megatama Indonesia:

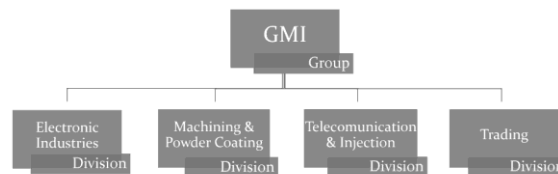
Site	Address / Role
Registered office	Plaza Summarecon Bekasi, Jl. Bulevar Ahmad Yani Kav. K.01, Level 7, Bekasi City, West Java
Production site 1	Jl. Kp. Jatipilar, Cikarang Selatan, Bekasi Regency
Production site 2	Jl. WR Supratman, Mustikajaya, Bekasi City

5) ISO 9001 certificate, quality manual, and procedure documents

Requirement: (If applicable) ISO 9001 certificate, quality manual, and directory of procedure documents (for QAR/QAN review).

ISO 9001 certification is currently in progress, with the audit planned for October 2026. The certificate, quality manual, and procedure-document index are pending submission.

Status: Open — certification and supporting documents in progress.



Type Machine	Spec	Year
CNC Late	Chuck 6 Inch	2024
CNC Z-Mate	Chuck 6 Inch	2025
CNC Milling	Table 1000 x 600	2024



Tonnage	Unit	Brand	L/T tie Bar
110	1	NPC	580 x 580 mm
160	1	NPC	580 x 580 mm
260	1	NPC	580 x 580 mm
360	1	NPC	670 x 670 mm
450	1	NPC	770 x 770 mm
650	1	NPC	870 x 870 mm



2. TECHNICAL DOCUMENTATION

1) Detailed product description

The Gas Leak Detector (GLD) is an IoT-based, multi-sensor gas-leak detection device intended for early detection and identification of flammable and selected toxic/process gases in oil and gas refinery environments, including process units, tank farms, pipe racks, and storage/loading-unloading areas. This document addresses the GLD unit itself as the subject of the current IECEx/ATEX certification.

Functionally, the GLD integrates eight metal-oxide-semiconductor (MOS) gas sensors of the MQ series, an ESP32-S3 microcontroller for local signal processing and edge-AI inference, and a LoRa radio module for wireless communication. The GLD is installed as a fixed-point detector at locations where gas leakage may occur.

The eight MQ sensors are simultaneously exposed to the sampled ambient gas. Their response patterns are acquired and processed locally by the ESP32-S3. An on-device AI gas-classification model analyzes the multi-sensor response so that gas-detection and classification functions do not depend on a continuous connection to a central server. When the sensor-response pattern and configured alarm criteria indicate a gas-leak condition, the GLD activates the local visual/audible alarm and transmits the alarm information over the LoRa network to the cluster head/gateway for forwarding to the operator monitoring system.

Ambient gas enters the sensing section through the stainless-steel mesh located at the front of the enclosure. During the sampling cycle, an internal DC fan assists gas flow into the sensing area where the eight MQ sensors are installed. Each MQ sensor incorporates an additional stainless-steel protective mesh surrounding its sensing element.

The GLD enclosure is intended for hazardous-area installation and is constructed primarily from aluminum alloy and stainless steel. The unit is mounted using a mechanical bracket to existing plant structures without requiring drilling or welding to the supporting structure.

The current production configuration of the GLD is powered continuously from an external 24 VDC supply. Where an AC/DC adapter is used to derive the 24 VDC supply from the site electrical system, the adapter is external to the GLD enclosure and is not part of the detector electronics covered by this product description.

2) Product name, model, and specification list

Item	Specification	Remarks
Product name	Gas Leak Detector (GLD)	Fixed-point multi-sensor gas detector
Model	GLD V2	Current certification model
Manufacturer	PT Galaksi Megatama Indonesia	
Primary function	Multi-sensor gas detection, local processing, alarm generation, and LoRa communication	—
Gas sensor configuration	8 channels: MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135	MOS gas sensors
Processing unit	ESP32-S3-WROOM-1U-N16R8	Edge processing / AI inference
Wireless communication	LoRa, E22-900MM22S	Star-topology communication
Operating frequency	920-923 MHz	
RF transmit power	0–22 dBm	Firmware-configurable
Main power input	24 VDC	Continuous production configuration
Internal supply rails	5 VDC and 3.3 VDC	5 V for sensor/heater circuitry; 3.3 V for logic
Maximum power consumption	8 W @ 24 VDC	
Local alarm	Visual and audible alarm	Integrated alarm module
Enclosure material	Aluminum alloy + stainless steel	—
Dimensions	200 × 90 × 290 mm	L × W × H
Weight	2378 g	
Mounting method	Bracket plate	Fixed installation
Cable entry	PG13.5	Needs confirmation against actual mechanical drawing / final approved entry
Antenna	External 3 dBi omnidirectional antenna, SMA interface	
Ambient temperature	–20 °C to +85 °C	Current draft value
Operating humidity	5–95% RH, non-condensing	Current draft value
Ingress protection target	IP66	Current draft states IP66; final certification status should follow actual test evidence
Intended hazardous area	Zone 1, Group II, proposed EPL Gb	Still pending ExCB confirmation
Gas group	IIC — proposed	Pending ExCB confirmation
Temperature class	T4 — proposed	Pending temperature evaluation/test
Protection concept	To be confirmed	Draft still shows Ex [db/ib] pending final decision

3) Functional description and technical parameters

Requirement: Complete and clear functional description and technical parameters (electrical parameters, mechanical parameters, etc.);

3.1 Functional description

The Gas Leak Detector (GLD) is a fixed-point gas detection unit that continuously monitors the surrounding atmosphere using an array of eight metal-oxide-semiconductor (MOS) gas sensors of the MQ series.

Ambient gas enters the sensing section through the stainless-steel mesh located at the front of the enclosure. During the sampling cycle, an internal DC fan is activated to assist gas flow through the front mesh and into the sensing area. Eight MQ-series gas sensors are installed within the sensing area. Each MQ sensor incorporates an additional stainless-steel protective mesh surrounding its sensing element.

The responses of all eight MQ sensors are acquired by the analog measurement circuitry and converted to digital data by the ADC. The sensor data are processed locally by the ESP32-S3 microcontroller. An edge-AI gas-classification algorithm evaluates the multi-sensor response pattern and determines the detected gas condition.

The detection and classification process is performed locally within the GLD and does not require a continuous connection to the central monitoring system.

When the configured alarm criteria indicate a gas-leak condition, the GLD activates the integrated local visual/audible alarm and transmits the alarm and detection information through the LoRa wireless communication link to the cluster head/gateway for forwarding to the operator monitoring system.

Under normal operation, the functional sequence is:

Ambient gas → front stainless-steel mesh → sampling fan → sensing chamber → MQ sensor mesh → MQ sensor array → analog acquisition → ADC → ESP32-S3 processing / AI inference → local alarm and LoRa transmission.

3.2 Gas sampling and sensor operation

The GLD uses eight MQ-series MOS sensors:

MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8 and MQ-135.

Ambient gas is introduced into the sensing area through the front stainless-steel mesh. The internal DC fan operates during the sampling interval to improve gas circulation through the sensing region.

The sampled gas subsequently passes through the integrated stainless-steel protective mesh of each MQ sensor before reaching the sensing element. The MQ sensing elements use internally heated metal-oxide sensing structures whose electrical responses vary according to the surrounding gas composition.

The eight sensor responses are evaluated collectively by the embedded processing algorithm rather than relying on a single MQ sensor as the sole indication of gas type.

3.3 Signal acquisition and edge-AI processing

The analog responses from the eight MQ sensor channels are acquired by the GLD analog front-end and digitized using an ADS1256IDBR 24-bit analog-to-digital converter.

The digitized sensor data are processed by an ESP32-S3-WROOM-1U-N16R8 microcontroller. The ESP32-

S3 performs sensor-data normalization, local signal processing, gas-pattern classification, alarm evaluation and communication control.

An embedded AI model operates locally on the ESP32-S3 so that the principal gas-detection and classification functions remain available without continuous rfd\7 communication with the external monitoring system.

3.4 Alarm operation

When the processed sensor response meets the configured alarm criteria, the GLD initiates two alarm paths:

- activation of the integrated local visual/audible alarm; and
- transmission of the alarm information over the LoRa wireless link to the cluster head/gateway.

The LoRa communication link is supplementary to the local alarm function; the gas-detection and classification process is performed locally by the GLD.

3.5 Wireless Communication

Parameter	Specification
Communication method	LoRa
Module	E22-900MM22S
Network topology	Star topology
Operating frequency	920–923 MHz
Nominal configured frequency	920 MHz
Transmit power	0–22 dBm, firmware configurable
Bandwidth	125 kHz
Spreading factor	SF7
Coding rate	4/5
Antenna	External omnidirectional antenna
Antenna gain	3 dBi
RF interface	SMA

3.6 Electrical Parameters — Node Sensor (GLD)

Parameter	Specification	Remarks
Rated power input	24 VDC	Continuous production configuration
External power source	External 24 VDC supply	AC/DC adapter external to GLD, if applicable
Internal sensor/heater rail	5 VDC	MQ sensing/heater circuitry
Internal logic rail	3.3 VDC	MCU and digital electronics
Maximum measured power consumption	8 W @ 24 VDC	Current draft value
Calculated current at 24 V / 8 W	Approx. 0.33 A	Not necessarily worst-case startup current
Input overvoltage/transient protection	33 V TVS	Current design
Overcurrent protection	Resettable PTC	Rating to be verified against actual circuit
Reverse-polarity/input protection	P-channel MOSFET stage	Current design

3.7 Sensor and measurement parameters

Function	Component / Specification
Gas sensor channels	8
Sensors	MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135
Sensor technology	Metal-oxide semiconductor
ADC	ADS1256IDBR
ADC resolution	24 bit
Environmental sensor	SHT40-AD1B-R2
Environmental parameters	Temperature and relative humidity
Processing MCU	ESP32-S3-WROOM-1U-N16R8

3.8 Wired communication interface

Parameter	Specification
Interface	RS-485 / Modbus RTU
Transceiver	THVD1410DR
Mode	Slave, read-only
Baud rate	9600 bit/s
Format	8N1

3.9 Mechanical Parameters — Node Sensor (GLD)

Parameter	Specification	Status / Remarks
Enclosure material	Aluminum alloy + stainless steel	Production construction
Dimensions	200 × 90 × 290 mm	Verify against controlled drawing
Weight	Approx. 2378 g	Current prototype/product value
Mounting method	Bracket plate	Fixed installation
Front gas inlet	Stainless-steel mesh assembly	See mechanical drawing
Internal sampling fan	DC fan	Operates during sampling
Sensor protective structure	Integrated stainless-steel mesh on each MQ sensor	Manufacturer construction
Cable entry	To be confirmed from controlled drawing	Draft currently says PG13.5
Antenna interface	SMA	External antenna
Ambient temperature	–20 °C to +85 °C	Current design target
Humidity	5–95% RH, non-condensing	Current draft
Ingress protection	IP66 target / as verified by test	Avoid calling Final before evidence

3.10 Operating modes

Mode	Function
Normal / Inference	Normal gas monitoring, AI processing and alarm function
Sampling	Fan activated and sensor data acquired
Alarm	Local visual/audible alarm and LoRa alarm transmission

Mode	Function
Nulling / Baseline	Sensor baseline operation in confirmed clean air
Dataset / Engineering	Engineering/data-collection mode; not normal production detection mode

4) Clear product photos (overall and key components)



Figure 4-1. GLD V2 – Complete assembled unit, front view.



Figure 4-2. GLD V2 – Complete assembled unit, Left view.

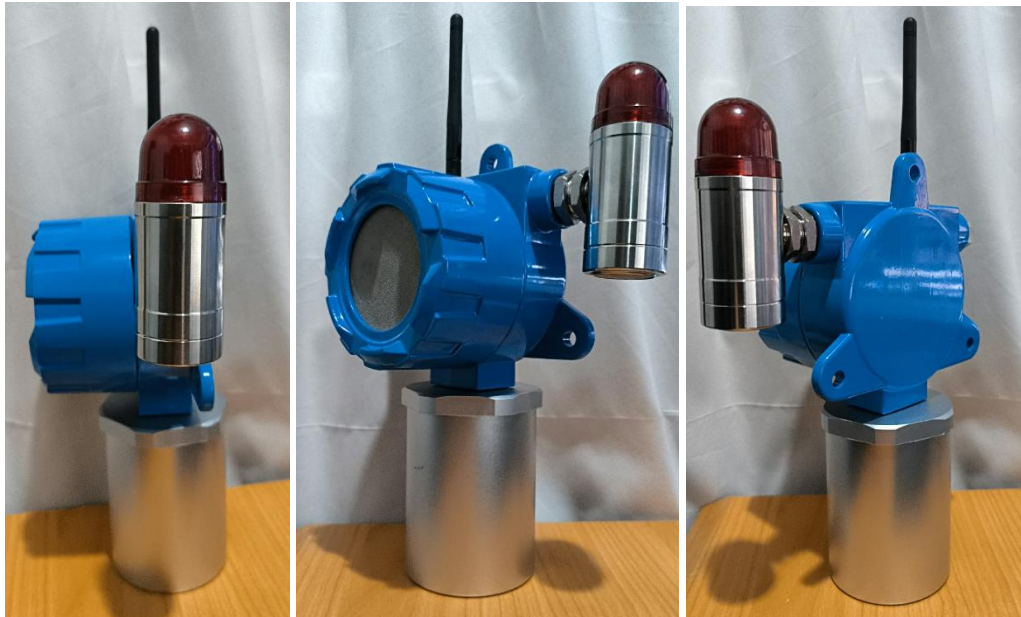


Figure 4-3. GLD V2 – Complete assembled unit, Right view.

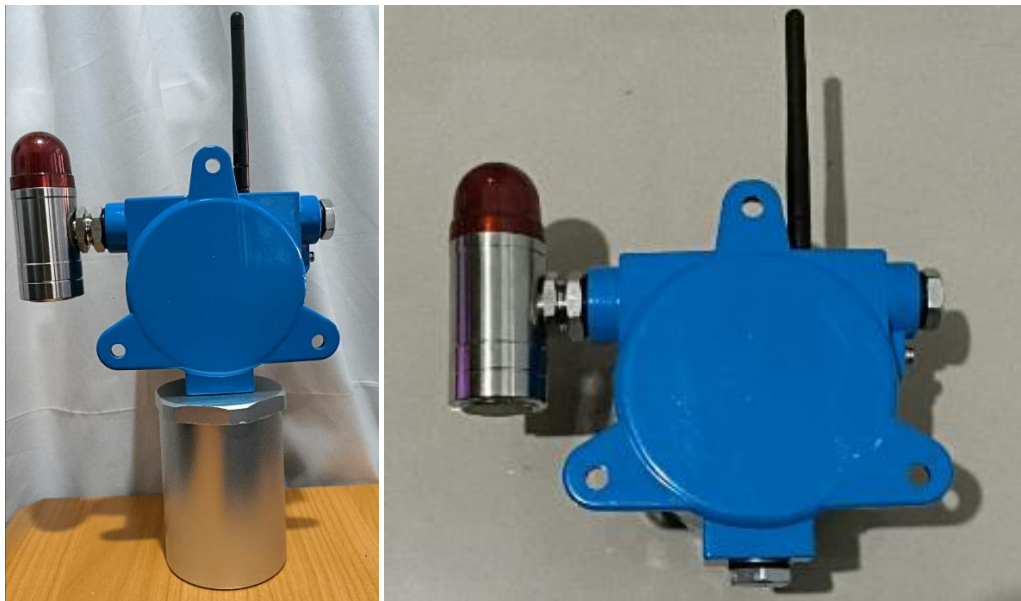


Figure 4-4. GLD V2 – Complete assembled unit, Back view.



Figure 4-5. GLD V2 – Cover with Stainless Wire-Mesh



Figure 4-6. GLD V2 – Base Casing GLD. Front View



Figure 4-7. GLD V2 – Base Casing GLD, Bottom View

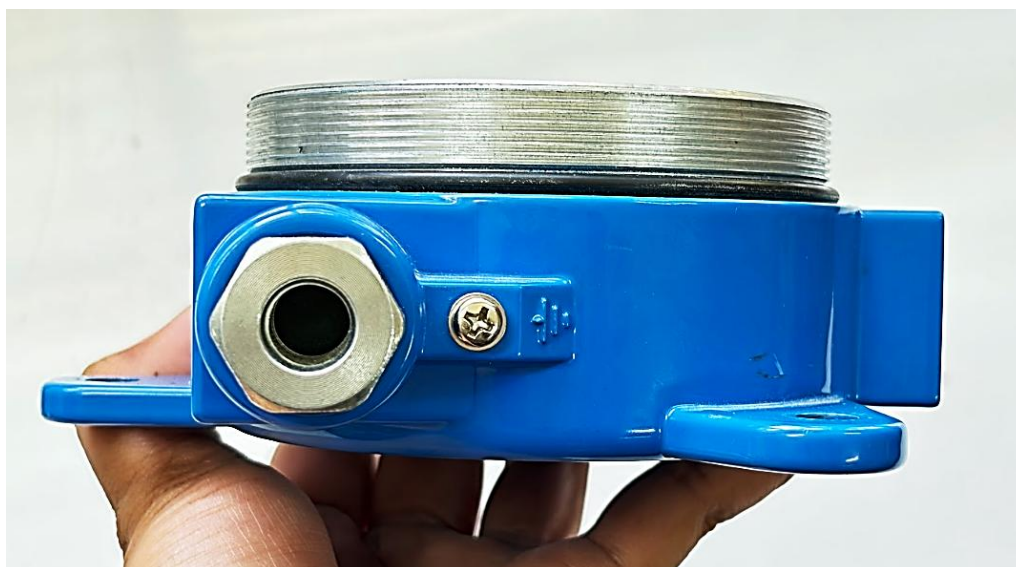


Figure 4-8. GLD V2 – Base Casing GLD, Left View



Figure 4-9. GLD V2 – Base Casing GLD, Top View



Figure 4-9. GLD V2 – Base Casing GLD, Right View

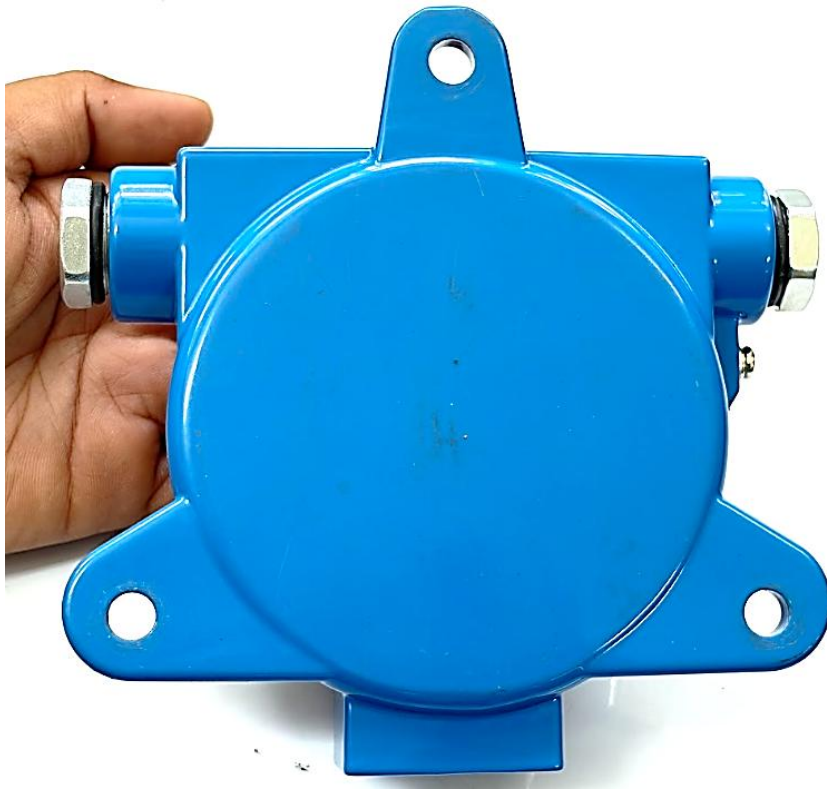


Figure 4-10. GLD V2 – Base Casing GLD, Back View



Figure 4-11. GLD V2 – Buzzer & LED, Top View

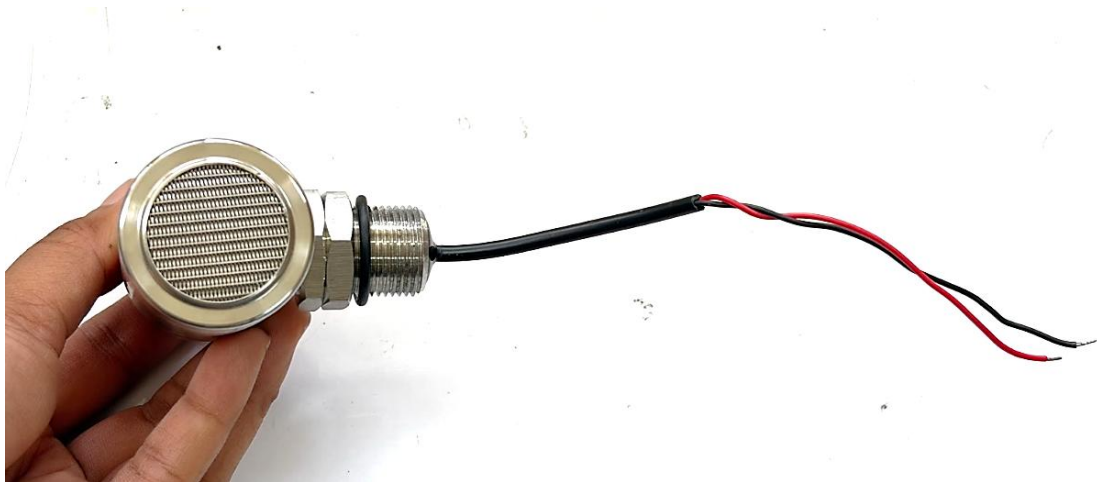


Figure 4-12. GLD V2 – Buzzer & LED, Bottom View



Figure 4-13. GLD V2 – Buzzer & LED, Side View



Figure 4-14. GLD V2 – LoRa Antenna

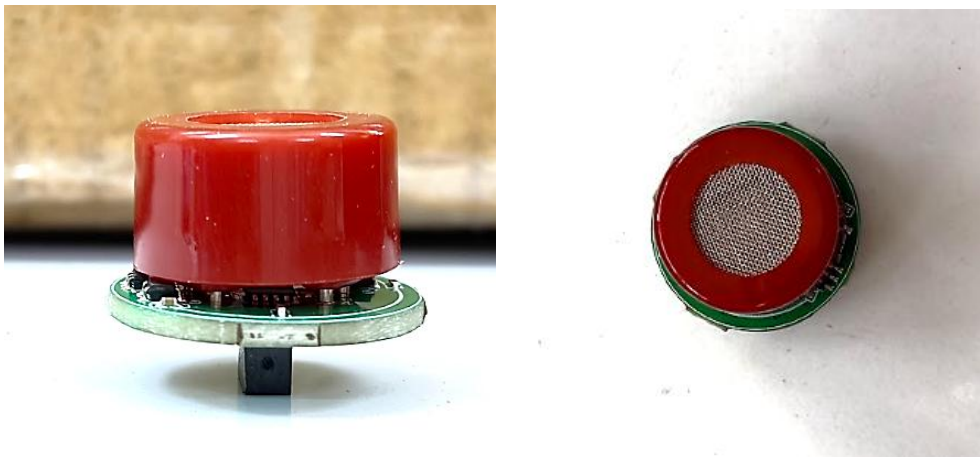


Figure 4-15. GLD V2 – MQ Sensor

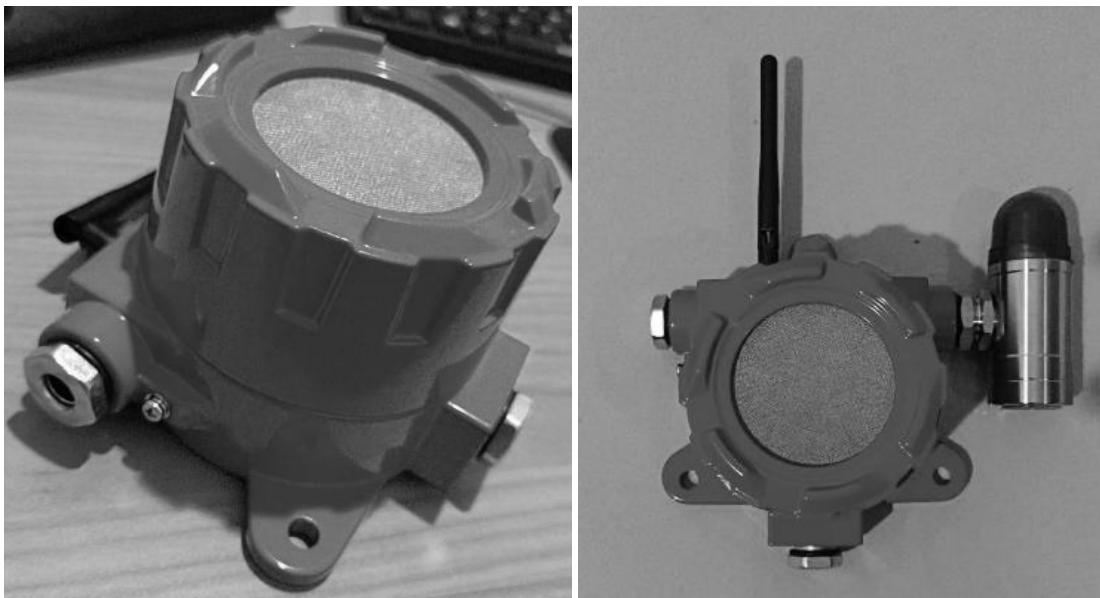


Figure 4-xx GLD V2 - Enclosure Body & Cover



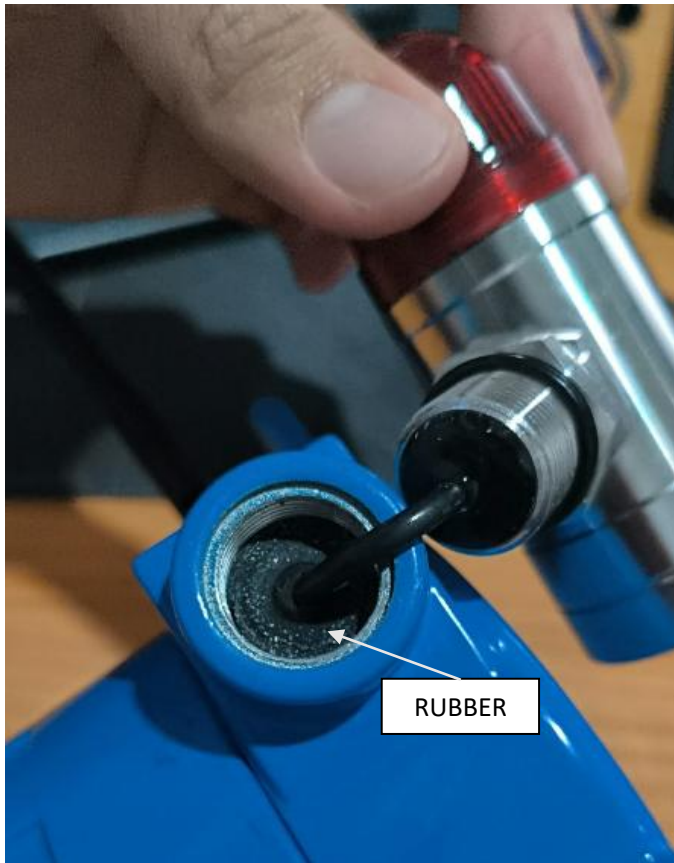


Figure 4-xx GLD V2 - Gasket - Rubber Gasket

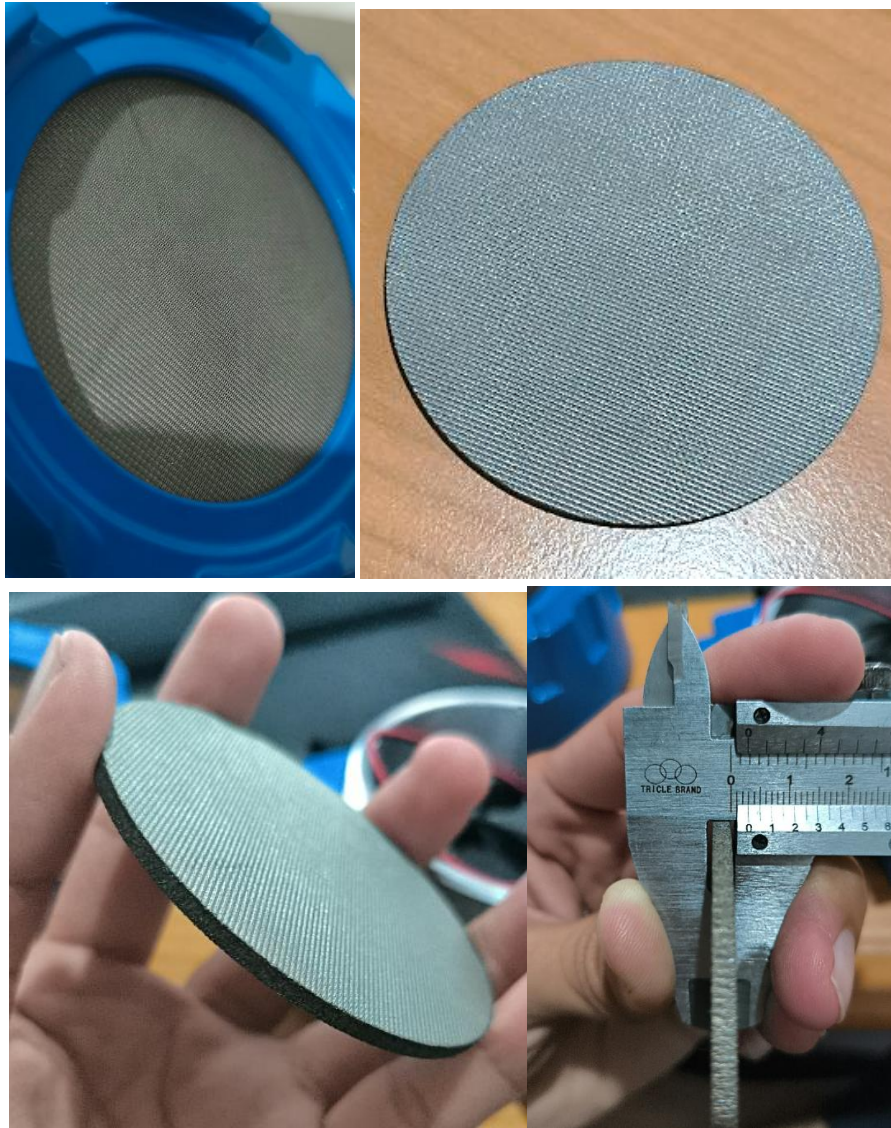


Figure 4-xx GLD V2 - Stainless Wire-Mesh Plate

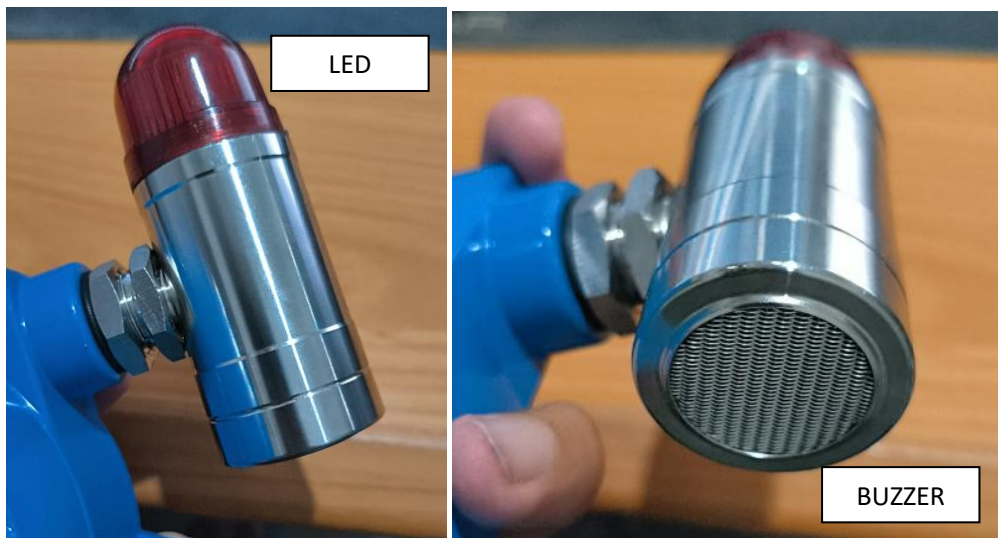


Figure 4-xx GLD V2 - Visual/Audible Alarm Beacon



Figure 4-xx GLD V2 – PCB Motherboard Inside Base Casing



Figure 4-xx GLD V2 – PCB Motherboard With MQ Sensors Inside Base Casing

5) Intended use and installation environment

Requirement: Description of intended use and installation environment (e.g., gas group IIC/IIB/IIA if applicable, temperature group T1-T6, ambient temperature range, area classification 0/1/2 or 20/21/22);

Intended Use

The Gas Leak Detector (GLD) is intended for continuous fixed-point monitoring of flammable and selected process gases in hazardous areas of oil and gas refinery facilities. Typical installation locations include process units, tank farms, pipe racks, and storage/loading-unloading areas. The detector is intended to provide local gas-leak detection and alarm indication and to transmit detection information to the remote monitoring system through the LoRa communication network.

Installation Environment

The GLD is intended for installation in outdoor industrial refinery environments where an explosive gas atmosphere may occur during normal operation. The current certification target is Equipment Group II, Category 2G / EPL Gb, for installation in Zone 1 gas hazardous areas.

Gas Group

The current proposed gas group is IIC, subject to confirmation by the ExCB. The product includes MQ-series sensors intended to respond to several gases, including hydrogen. The final gas-group classification shall be confirmed as part of the certification assessment.

Temperature Class

The current proposed temperature class is T4. Final assignment of the temperature class shall be based on the maximum surface-temperature evaluation and applicable test results.

Ambient Conditions

The intended ambient operating temperature range is –20 °C to +60 °C. The intended operating humidity range is 5–95% RH, non-condensing.

Installation Conditions

The GLD is intended for fixed installation using a mechanical mounting bracket. Electrical power is supplied from an external 24 VDC source. Installation, cable entries, grounding, antenna arrangement, and enclosure assembly shall comply with the approved product configuration and the requirements defined in the installation instructions.

Certification Scope

The current certification scope addresses hazardous gas atmospheres. Dust hazardous-area classification is not included in this section unless separately evaluated and approved.

6) Design and manufacturing information

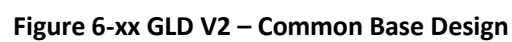
a. Complete drawings

Requirement: Complete drawings: Assembly drawing, component drawing, electrical schematic diagram, PCB layout (if applicable), enclosure structure drawing (indicating key explosion-proof parameters such as gap, length, volume), junction box drawing, terminal drawing, grounding connection drawing, etc. Drawings should clearly indicate dimensions, tolerances, and materials;

A controlled set of mechanical and electrical drawings is maintained for the GLD V2. The drawing package defines the complete product construction, including the enclosure, gas-sampling path, stainless-steel mesh assemblies, DC sampling fan, MQ sensor arrangement, motherboard PCB, alarm module, antenna interface, cable entry, terminals, grounding/bonding connections, and mounting arrangement.

All safety-critical dimensions, materials, tolerances, threads, joint lengths, internal volumes, sealing arrangements, and applicable explosion-protection-related construction parameters are specified on the corresponding controlled drawings. Each drawing is identified by drawing number, revision, title, and date.

IECEx/ATEX Certification Information Requirements – GLD V2



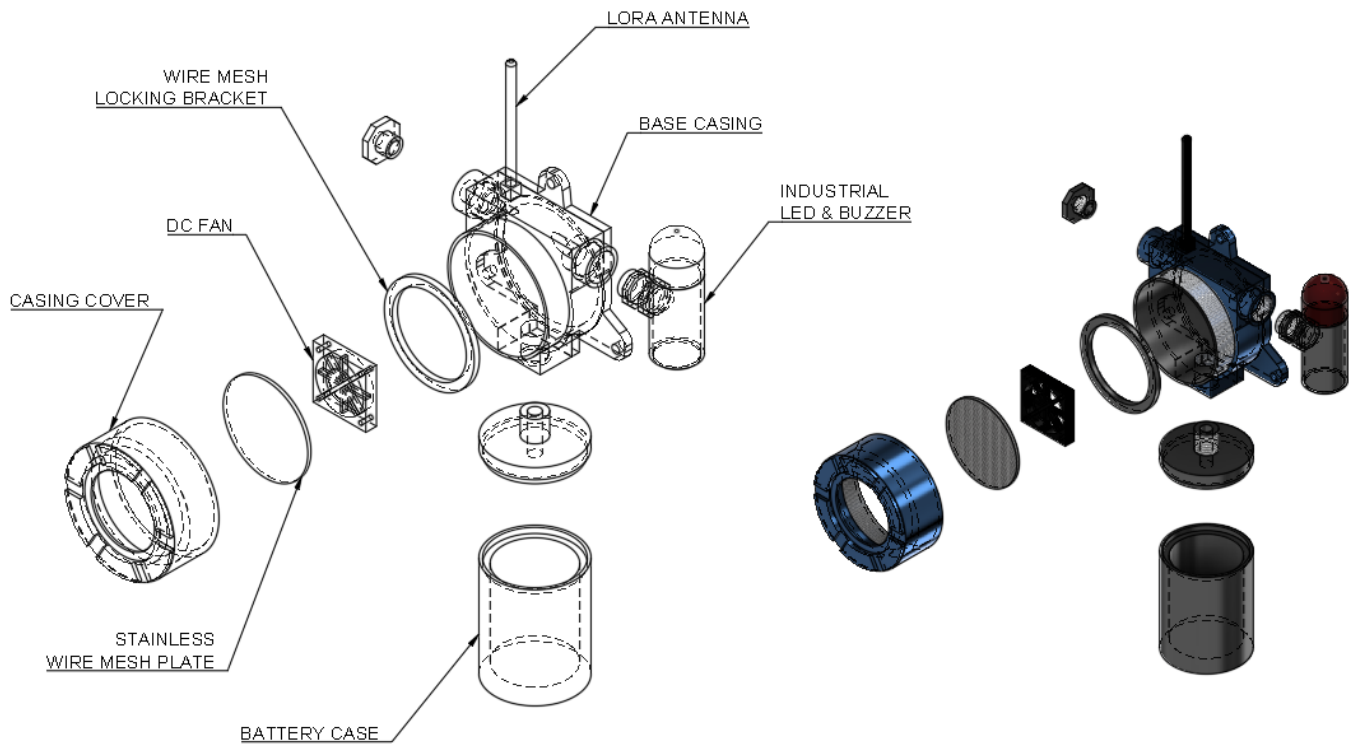


Figure 6-xx GLD V2 – Overall Exploded View I

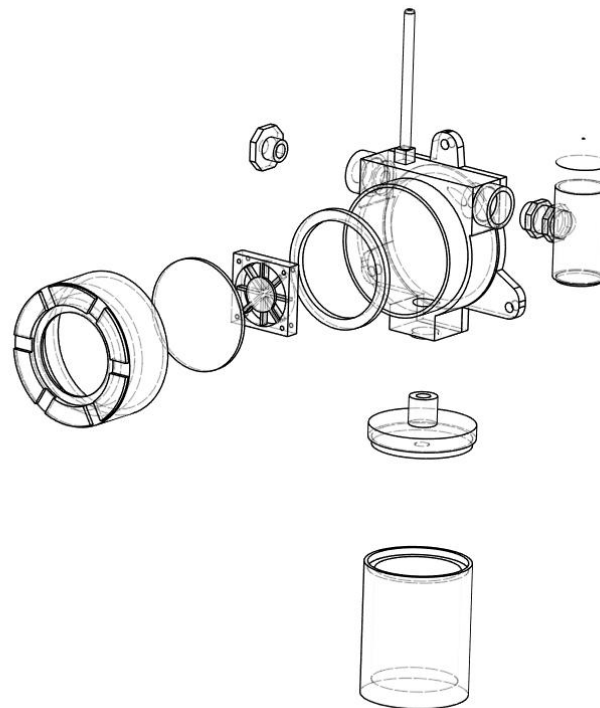


Figure 6-xx GLD V2 – Overall Exploded View II

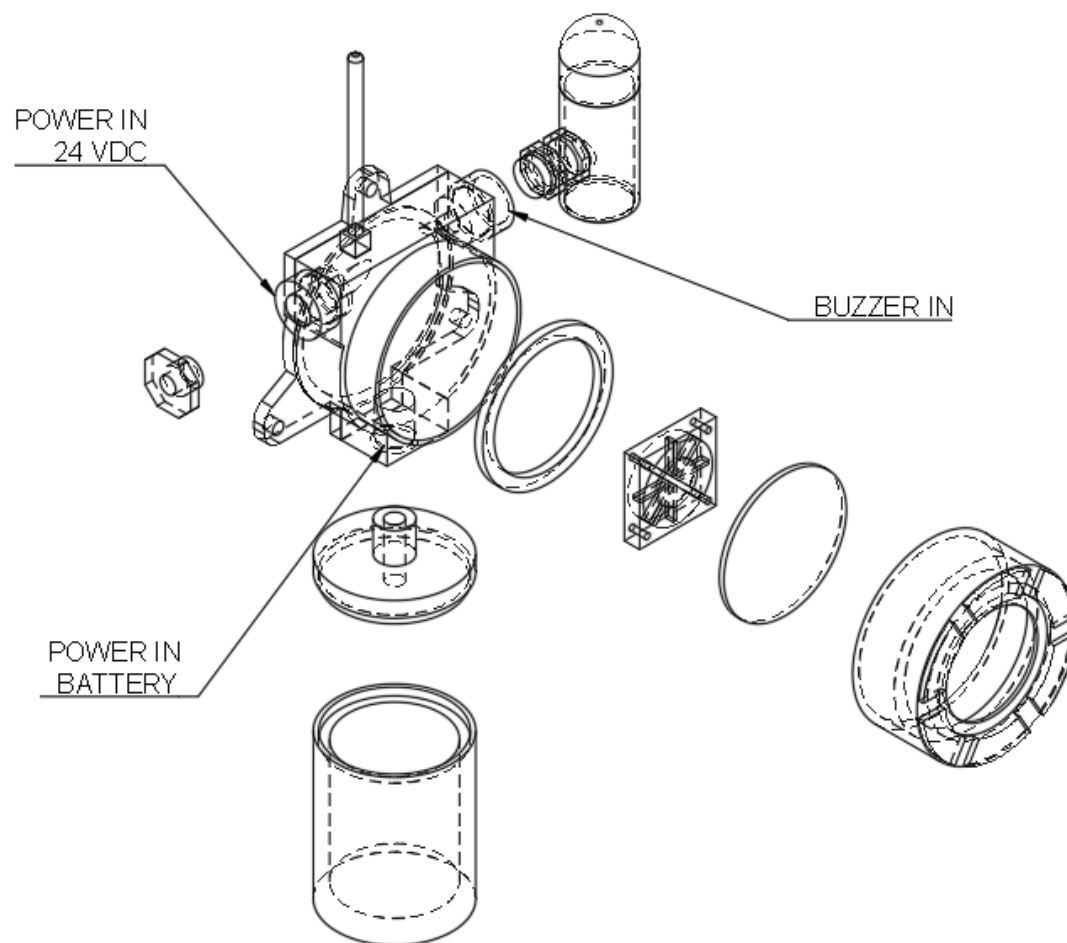


Figure 6-xx GLD V2 – Overall Exploded View III

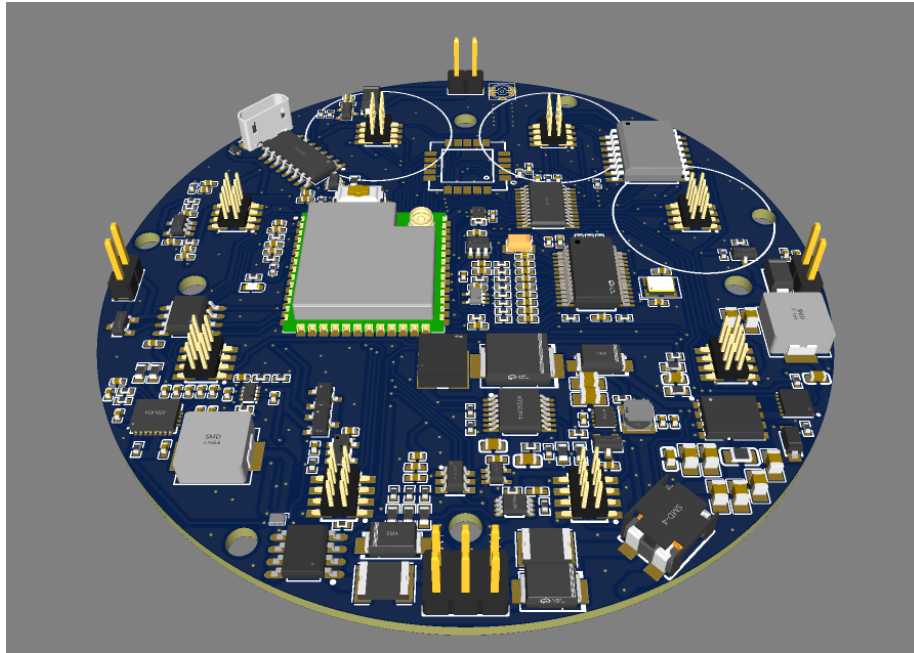


Figure 6-xx GLD V2 – MOTHERBOARD PCB

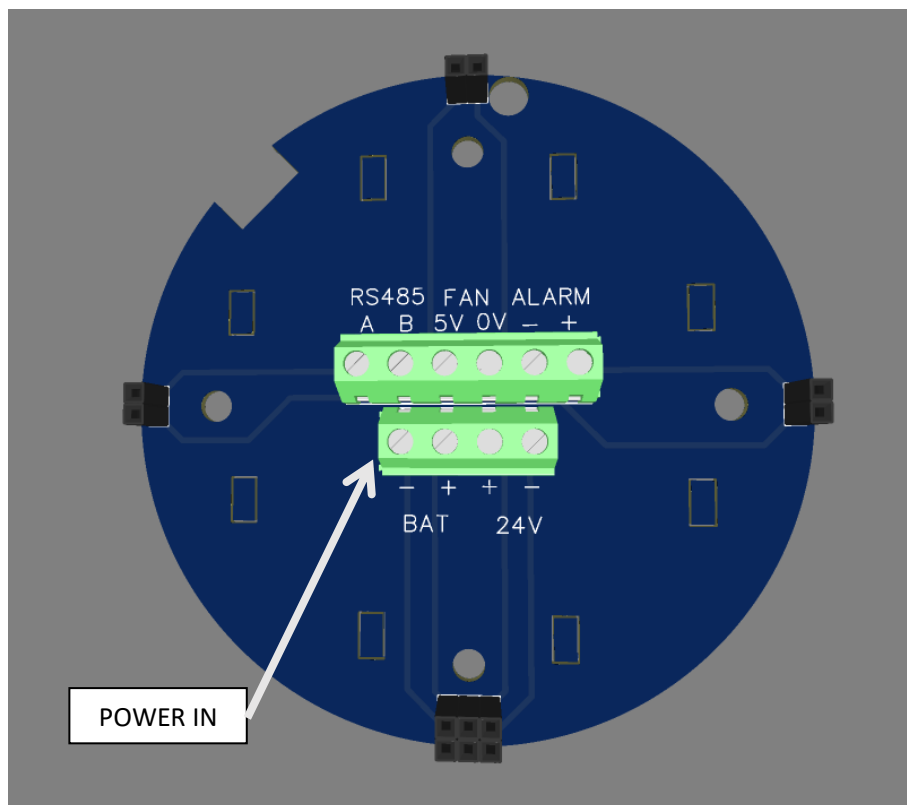
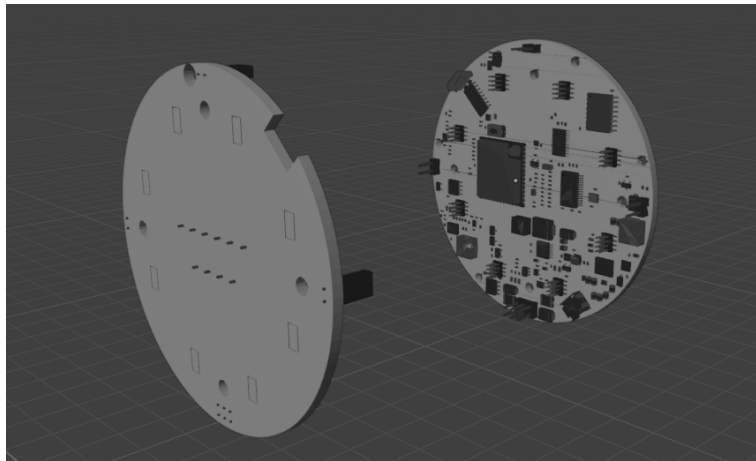


Figure 6-xx GLD V2 – PCB COVER (TERMINAL MODULE)



PCB COVER

MOTHERBOARD PCB

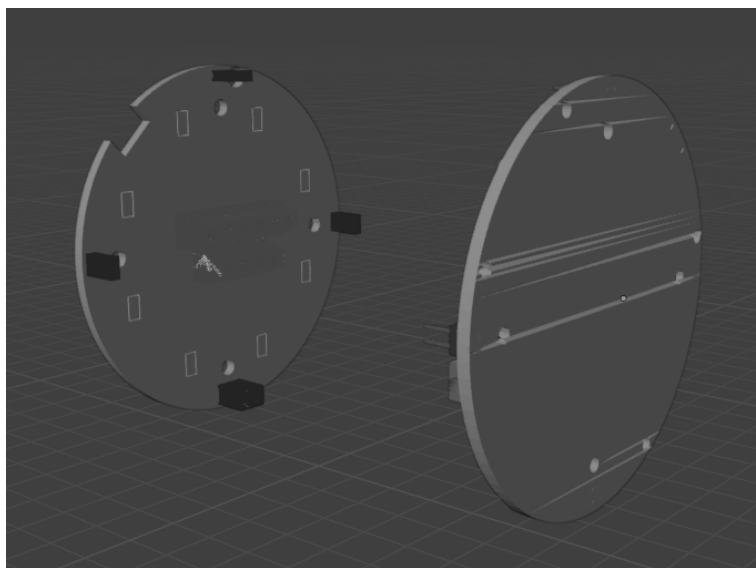
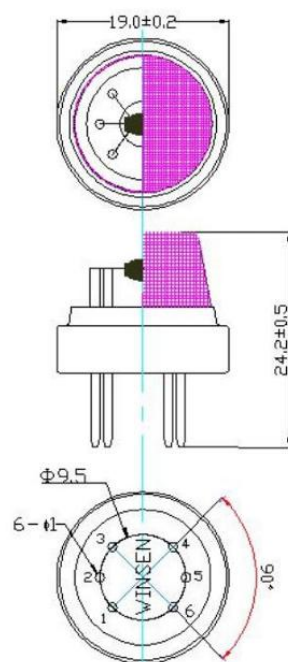


Figure 6-xx GLD V2 – Motherboard & PCB Cover

MQ Sensor Technical Parameters

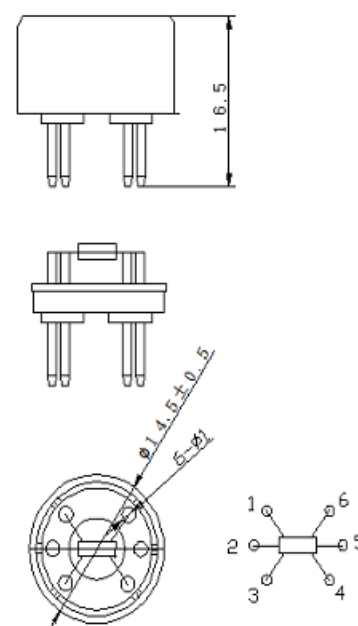
MQ-2

Model			MQ-2
Sensor Type			Semiconductor
Standard Encapsulation			Bakelite, Metal cap
Target Gas			Flammable gas, smoke
Detection range			300~10000ppm(flammable gas)
Standard Circuit Conditions	Loop Voltage	V_c	5.0V±0.1V DC
	Heater Voltage	V_H	5.0V±0.1V AC or DC
	Load Resistance	R_L	Adjustable
Sensor character under standard test conditions	Heater Resistance	R_H	30Ω±3Ω (room temp.)
	Heater consumption	P_H	≤950mW
	Sensitivity	S	$R_s(\text{in air})/R_s(2000\text{ppm C}_3\text{H}_8) \geq 5$
	Output Voltage	V_s	2.5V~4.0V (in 2000ppm C ₃ H ₈)
	Concentration Slope	α	$\leq 0.6(R_{3000\text{ppm}}/R_{1000\text{ppm C}_3\text{H}_8})$
Standard test conditions	Tem. Humidity		20℃±2℃; 55%±5%RH
	Standard test circuit		$V_c: 5.0V \pm 0.1V$ $V_H: 5.0V \pm 0.1V$
	Preheat time		Not less than 48 hours
	O2 content		21% (not less than 18%) O2 concentration effects initial value, sensitivity and repeatability.
Lifespan			10 years



MQ-3

Model			MQ-3
Sensor Type			Semiconductor
Standard Encapsulation			Plastic cap
Target Gas			Alcohol gas
Detection range			25~500ppm alcohol
Standard Circuit Conditions	Loop Voltage	V_c	≤24V DC
	Heater Voltage	V_H	5.0V±0.1V AC or DC
	Load Resistance	R_L	Adjustable
Sensor character under standard test conditions	Heater Resistance	R_H	29Ω±3Ω (room tem.)
	Heater consumption	P_H	≤900mW
	Sensitivity	S	$R_o(\text{in air})/R_s(125\text{ppm C}_2\text{H}_5\text{OH}) \geq 5$
	Output Voltage	V_s	2.5V~4.0V (in 125ppm C ₂ H ₅ OH)
	Concentration Slope	α	$\leq 0.6(R_{300\text{ppm}}/R_{50\text{ppm C}_2\text{H}_5\text{OH}})$
Standard test conditions	Tem. Humidity		20℃±2℃; 55%±5%RH
	Standard test circuit		$V_c: 5.0V \pm 0.1V$ $V_H: 5.0V \pm 0.1V$
	Preheat time		Over 48 hours



MQ-4

Model			MQ-4
Sensor Type			Semiconductor
Standard Encapsulation			Bakelite, Metal cap
Target Gas			Methane
Detection range			300~10000ppm(CH ₄)
Standard Circuit Conditions	Loop Voltage	V _c	5.0V±0.1V DC
	Heater Voltage	V _H	5.0V±0.1V AC or DC
	Load Resistance	R _L	Adjustable
Sensor character under standard test conditions	Heater Resistance	R _H	28 Ω ± 3 Ω (room temp.)
	Heater consumption	P _H	≤1W
	Sensitivity	S	Rs(in air)/Rs(in 5000ppmCH ₄)≥5
	Output Voltage	V _s	2.5V~4.0V (in 5000ppm CH ₄)
	Concentration Slope	α	≤0.6(R _{5000ppm} /R _{1000ppm} CH ₄)
Standard test conditions	Temp. Humidity	20℃±2℃; 55%±5%RH	
	Standard test circuit	V _c :5.0V±0.1V V _H :5.0V±0.1V	
	Preheat time	Not less than 48 hours	
	O ₂ content	21% (not less than 18%) O ₂ concentration effects initial value, sensitivity and repeatability.	
Lifespan			10 years

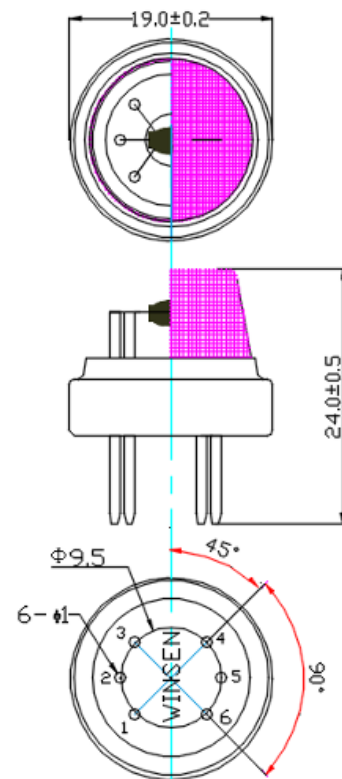


Fig1. Sensor Structure
Unit: mm Tolerance: ±0.1mm

MQ-5

Model			MQ-5
Sensor Type			Semiconductor
Standard Encapsulation			Bakelite, Metal cap
Target Gas			LPG, CH ₄
Detection range			300~10000ppm (CH ₄ , C ₃ H ₈)
Standard Circuit Conditions	Loop Voltage	V _c	≤24V DC
	Heater Voltage	V _H	5.0V±0.1V AC or DC
	Load Resistance	R _L	Adjustable
Sensor character under standard test conditions	Heater Resistance	R _H	26Ω±3Ω (room temp.)
	Heater consumption	P _H	≤950mW
	Sensitivity	S	Rs(in air)/Rs(in 2000ppm C ₃ H ₈)≥5
	Output Voltage	V _s	2.5V~4.0V (in 2000ppm C ₃ H ₈)
	Concentration Slope	α	≤0.6(R _{3000ppm} /R _{1000ppm} C ₃ H ₈)
Standard test conditions	Tem. Humidity	20℃±2℃; 55%±5%RH	
	Standard test circuit	V _c :5.0V±0.1V V _H :5.0V±0.1V	
	Preheat time	Not less than 48 hours	
	O ₂ content	21% (not less than 18%) O ₂ concentration effects initial value, sensitivity and repeatability.	
Lifespan			10 years

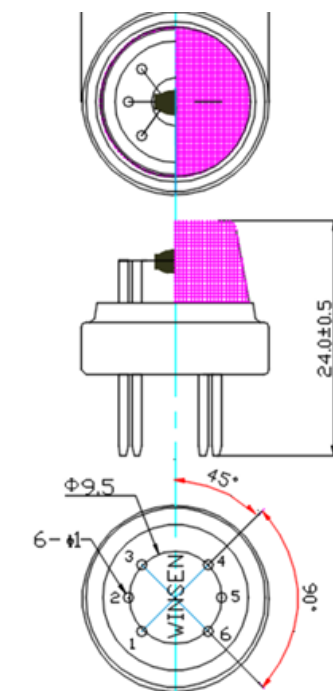


Fig1. Sensor Structure
Unit: mm Tolerance: ±0.1mm

MQ-6

Model			MQ-6
Sensor Type			Semiconductor
Standard Encapsulation			Bakelite, Metal cap
Target Gas			LPG
Detection range			300~10000ppm (Propane)
Standard Circuit Conditions	Loop Voltage	V_c	$\leq 24V$ DC
	Heater Voltage	V_H	$5.0V \pm 0.1V$ AC or DC
	Load Resistance	R_L	Adjustable
Sensor character under standard test conditions	Heater Resistance	R_H	$26\Omega \pm 3\Omega$ (room tem.)
	Heater consumption	P_H	$\leq 950mW$
	Sensitivity	S	$R_s(\text{in air})/R_s(\text{in } 2000ppm \text{ } C_3H_8) \geq 5$
	Output Voltage	V_s	$2.5V \sim 4.0V$ (in 2000ppm C_3H_8)
	Concentration Slope	α	$\leq 0.6(R_{3000ppm}/R_{1000ppm} \text{ } C_3H_8)$
Standard test conditions	Tem. Humidity		$20^\circ C \pm 2^\circ C$; $55\% \pm 5\% RH$
	Standard test circuit		$V_c: 5.0V \pm 0.1V$; $V_H: 5.0V \pm 0.1V$
	Preheat time		Over 48 hours

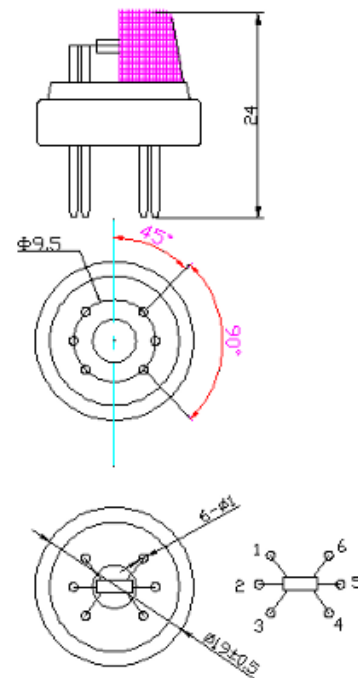


Fig1. Sensor Structure

MQ-7

Model			MQ-7
Sensor Type			Semiconductor
Standard Encapsulation			Plastic cap
Target Gas			carbon monoxide
Detection range			10~500ppm CO
Standard Circuit Conditions	Loop Voltage	V_c	$\leq 10V$ DC
	Heater Voltage	V_H	$5.0V \pm 0.1V$ AC or DC (High tem.) $1.5V \pm 0.1V$ AC or DC (Low tem.)
	Heater Time	T_L	$60 S \pm 1S$ (High tem.), $90 S \pm 1S$ (Low tem.)
	Load Resistance	R_L	Adjustable
Sensor character under standard test conditions	Heater Resistance	R_H	$29\Omega \pm 3\Omega$ (room tem.)
	Heater consumption	P_H	$\leq 900mW$
	Sensitivity	S	$R_s(\text{in air})/R_s(\text{in } 150ppm \text{ CO}) \geq 5$
	Output Voltage	V_s	$2.5V \sim 4.3V$ (in 150ppm CO)
Standard test conditions	Concentration Slope	α	$\leq 0.6(R_{300ppm}/R_{50ppm} \text{ CO})$
	Tem. Humidity		$20^\circ C \pm 2^\circ C$; $55\% \pm 5\% RH$
	Standard test circuit		$V_c: 5.0V \pm 0.1V$; V_H (High tem.): $5.0V \pm 0.1V$; V_H (Low tem.): $1.5V \pm 0.1V$
Standard test conditions	Preheat time		Over 48 hours

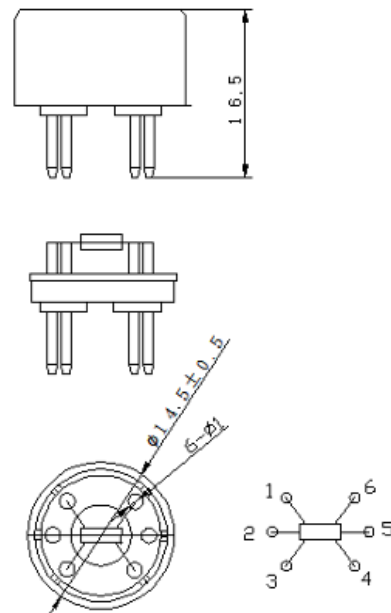


Fig1. Sensor Structure

Unit: mm

MQ-8

Model			MQ-8
Sensor Type			Semiconductor
Standard Encapsulation			Bakelite, Metal cap
Target Gas			Hydrogen
Detection range			100~1000ppm(H ₂ gas)
Standard Circuit Conditions	Loop Voltage	V _c	≤24V DC
	Heater Voltage	V _H	5.0V±0.1V AC or DC
	Load Resistance	R _L	Adjustable
Sensor character under standard test conditions	Heater Resistance	R _H	29Ω±3Ω(room tem.)
	Heater consumption	P _H	≤900mW
	Sensitivity	S	Rs(in air)/Rs(in 1000ppm H ₂)≥5
	Output Voltage	V _s	2.5V~4.0V (in 1000ppm H ₂)
	Concentration Slope	α	≤0.6(R _{1000ppm} /R _{400ppm} H ₂)
Standard test conditions	Tem. Humidity	20℃±2℃; 55%±5%RH	
	Standard test circuit	V _c :5.0V±0.1V; V _H :5.0V±0.1V	
	Preheat time	Over 48 hours	

NOTE: Output voltage (Vs) is V_{RL} in test environment.

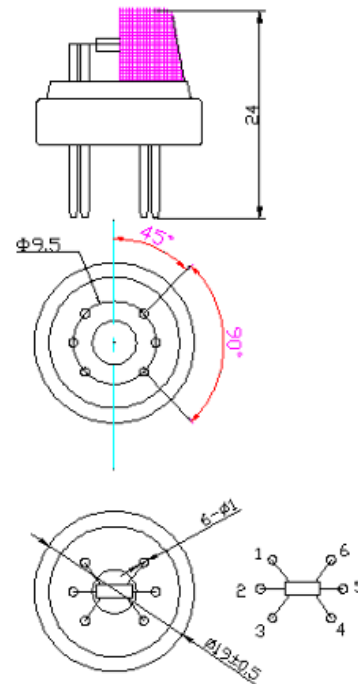


Fig1. Sensor Structure
Unit: mm

MQ-135

Model			MQ135
Sensor Type			Semiconductor
Standard Encapsulation			Bakelite, Metal cap
Target Gas			ammonia gas, sulfide, benzene series steam
Detection range			10~1000ppm(ammonia gas, toluene, hydrogen, smoke)
Standard Circuit Conditions	Loop Voltage	V _c	≤24V DC
	Heater Voltage	V _H	5.0V±0.1V AC or DC
	Load Resistance	R _L	Adjustable
Sensor character under standard test conditions	Heater Resistance	R _H	29Ω±3Ω (room tem.)
	Heater consumption	P _H	≤950mW
	Sensitivity	S	Rs(in air)/Rs(in 400ppm H ₂)≥5
	Output Voltage	V _s	2.0V~4.0V (in 400ppm H ₂)
	Concentration Slope	α	≤0.6(R _{400ppm} /R _{100ppm} H ₂)
Standard test conditions	Tem. Humidity	20℃±2℃; 55%±5%RH	
	Standard test circuit	V _c :5.0V±0.1V; V _H : 5.0V±0.1V	
	Preheat time	Over 48 hours	

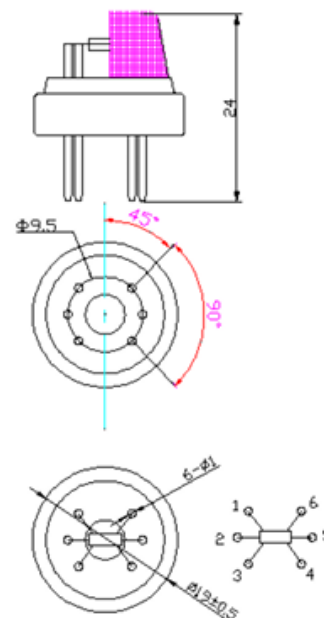
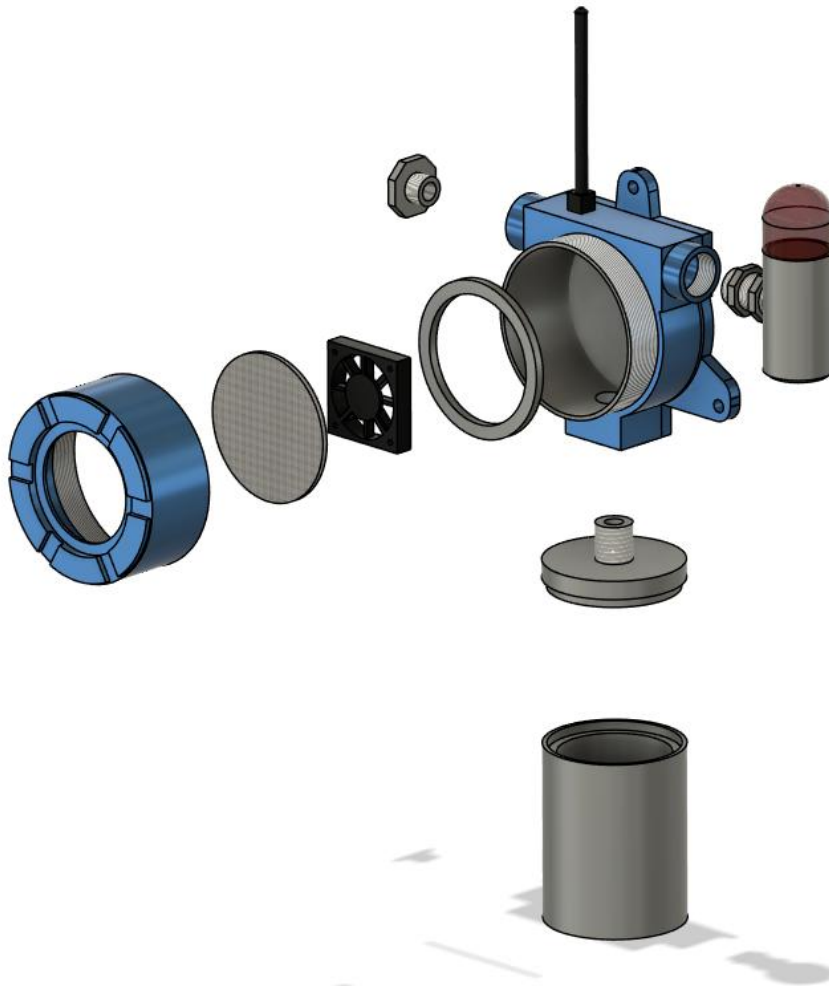
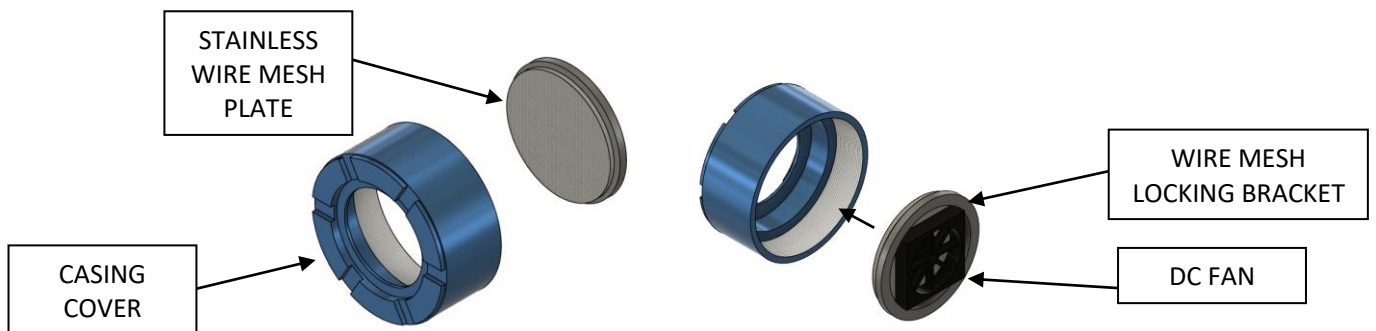


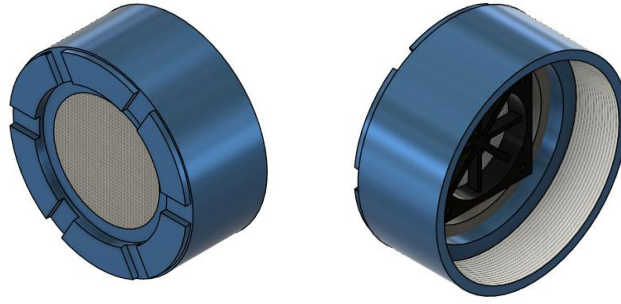
Fig1. Sensor Structure
Unit: mm

Assembly Drawing

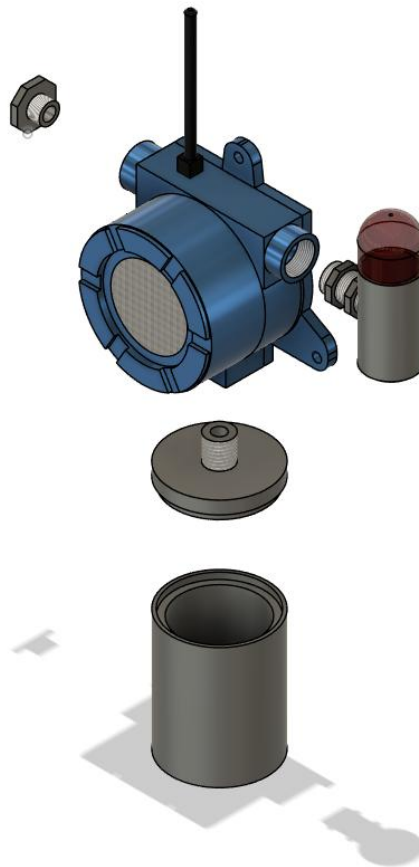


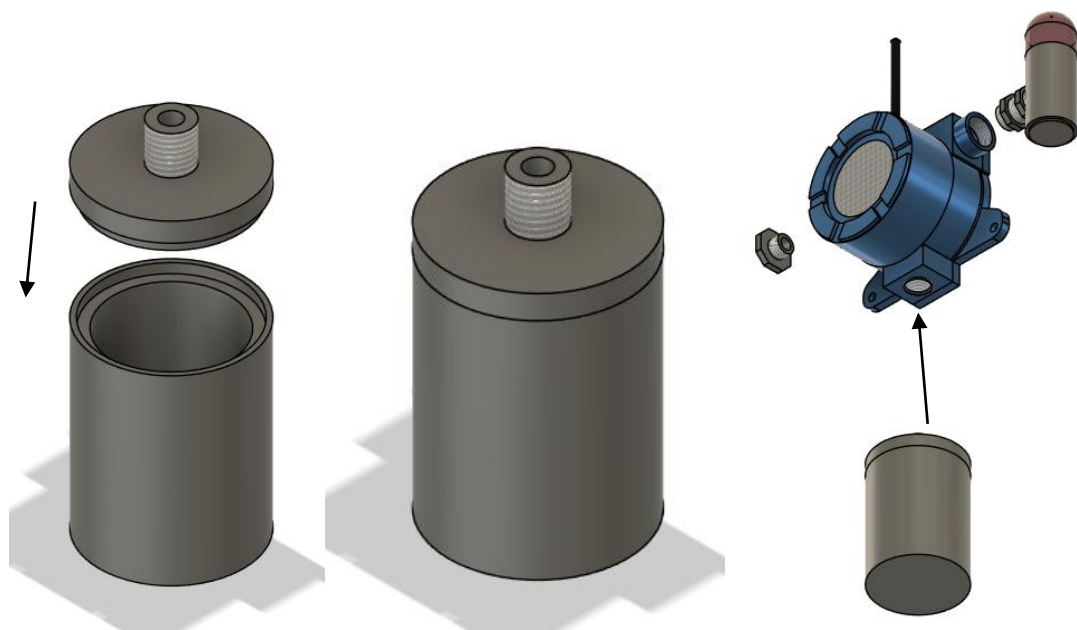
EXPLODED VIEW



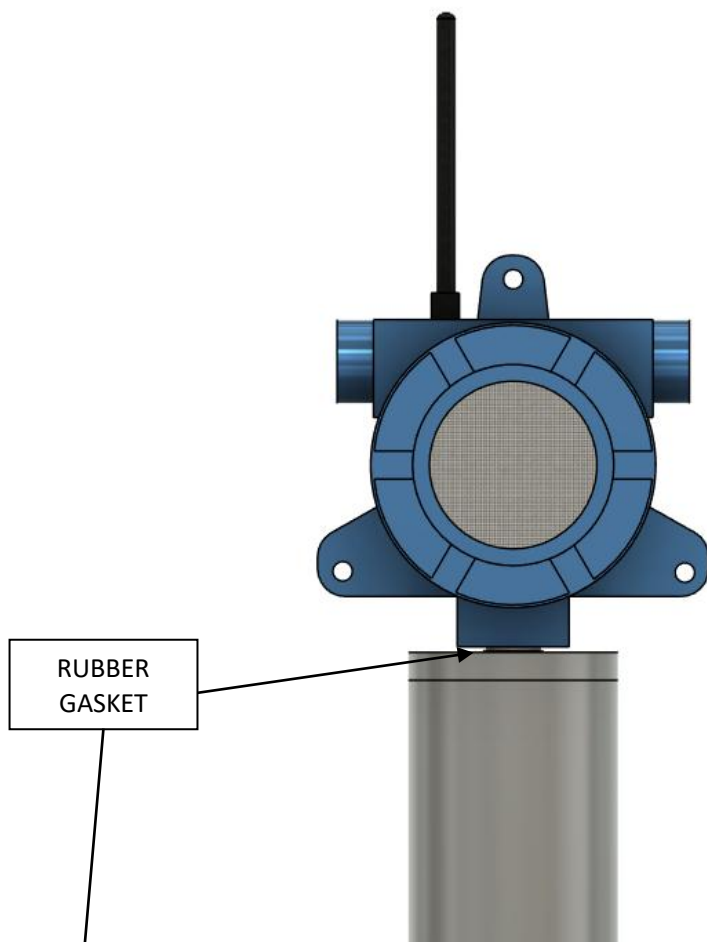


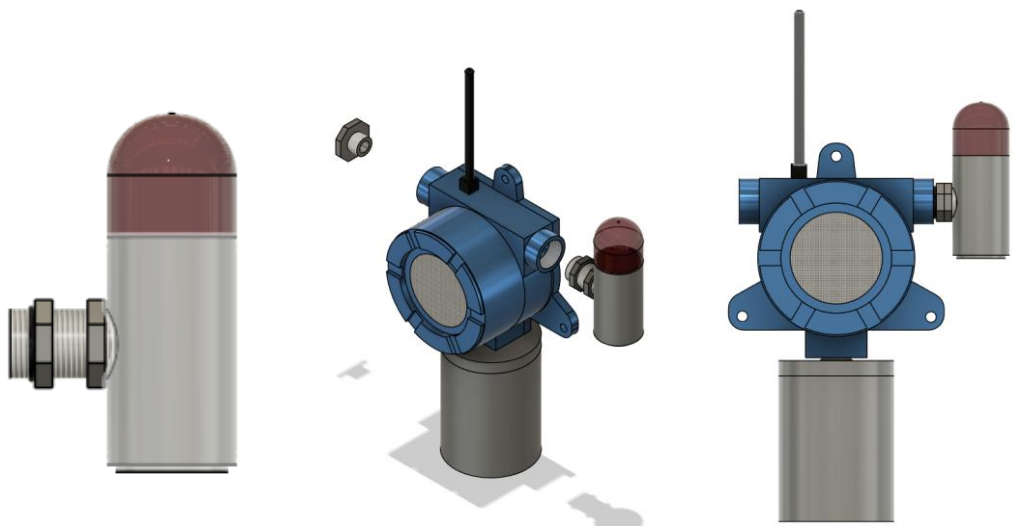
CASING COVER



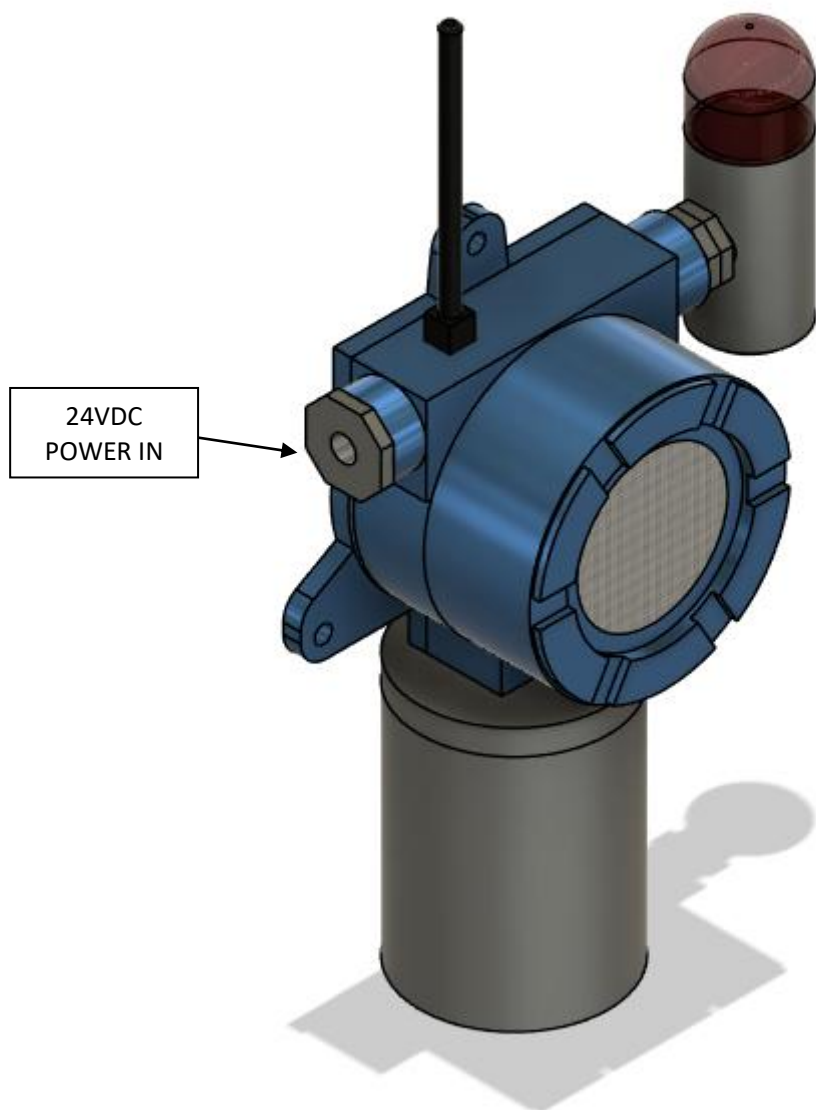


BATTERY CASE





LED & BUZZER





BASE CASING



PUT MOTHERBOARD TO BASE CASE

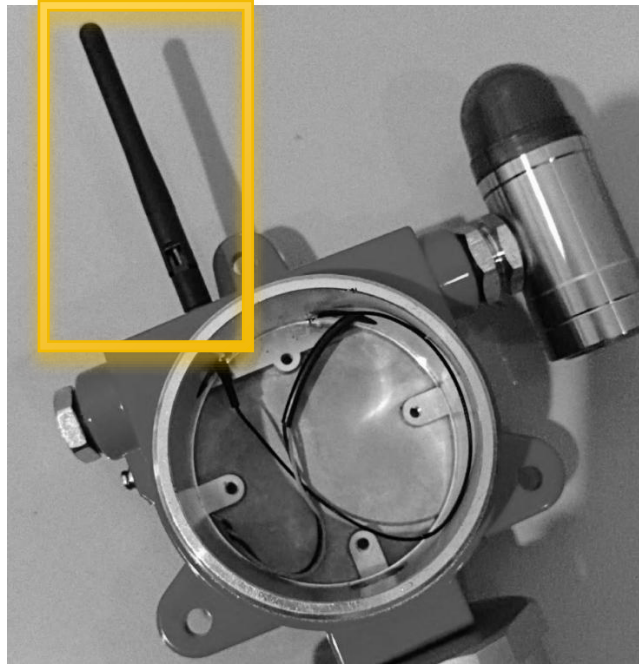


SECURE THE MOTHERBOARD TO THE BASE ENCLOSURE USING THE DESIGNATED MOUNTING SCREWS.



INSTALL LORA ANTENNA CABLE.

Note: In the latest version, the LoRa antenna is positioned above the base enclosure rather than on the side. The version currently shown reflects the previous design.



the latest version, the LoRa antenna is positioned above the base enclosure



CONNECT LED AND BUZZER CABLES.



**ATTACH DC FAN TO THE ENCLOSURE COVER,
BEHIND STAINLESS-STEEL WIRE-MESH PLATE.**

CONNECT DC FAN POWER CABLE TO MOTHERBOARD.



CLOSE THE ENCLOSURE COVER BY TURNING IT CLOCKWISE.

Component Drawing



ENCLOSURE



PCB + SENSOR



COVER + FAN



ALARM MODULE



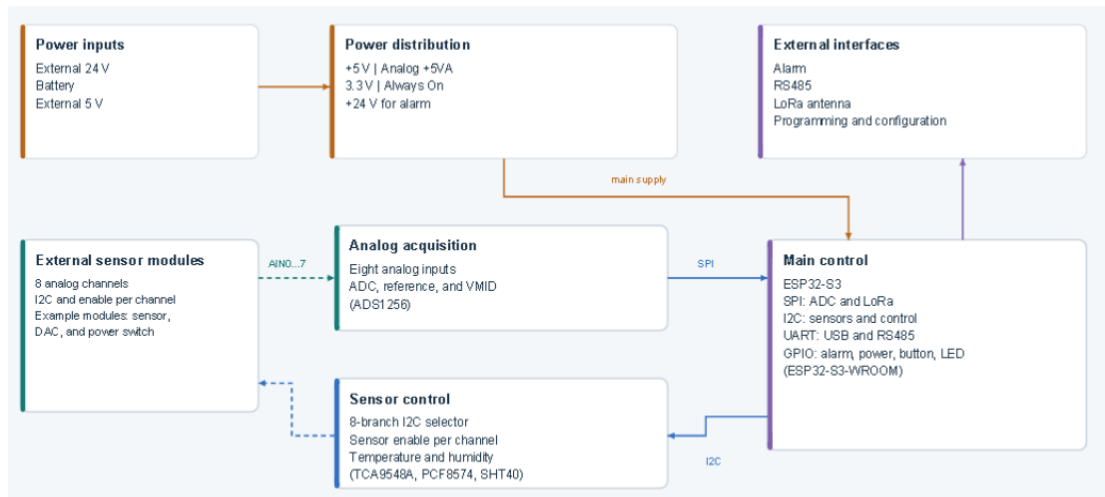
COVER + MESH



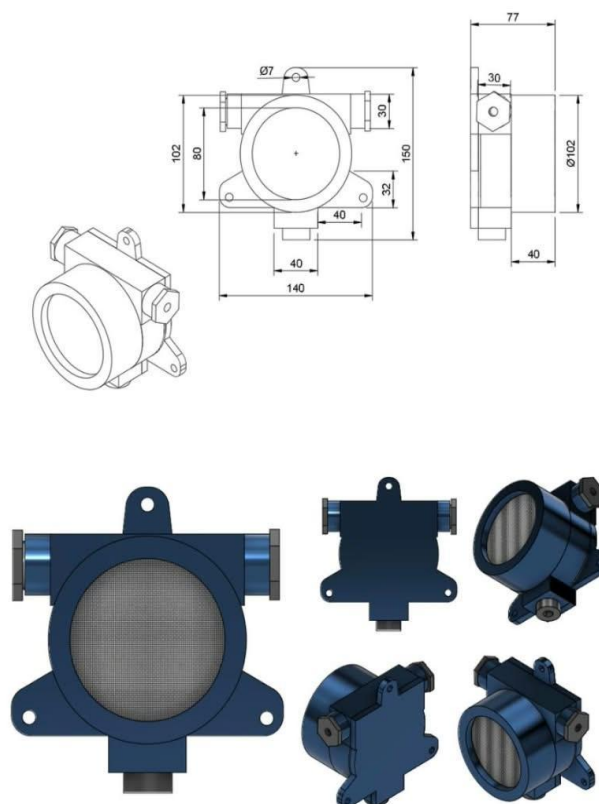
ANTENNA 3 DBI

Electrical Schematic

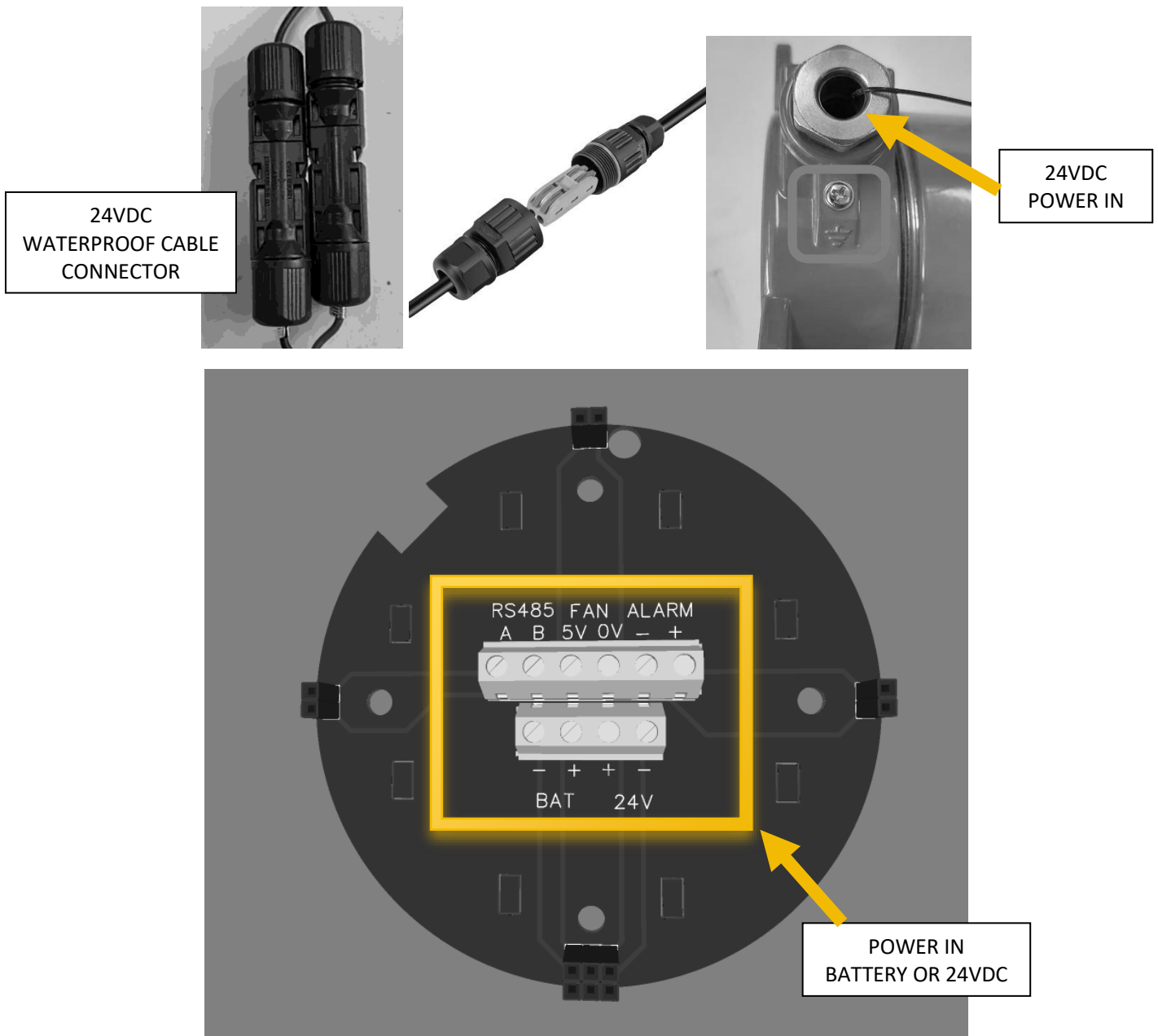
An electrical schematic capture and a corresponding PCB layout exist for the Gas Leak Detector Main board/ Motherboard .



Enclosure Structure



Terminal



Grounding



External grounding stud, marked with the protective earth (ground) symbol, installed on the enclosure body.

b. Bill of Materials (BOM) for key components

Requirement: Bill of Materials (BOM) for key components: List all components that affect explosion-proof safety (such as enclosures(gaskets, terminals, cable entry devices, switches, light sources, batteries, potting compounds, plastic parts, printed circuit boards, etc.), including manufacturer, model and specification, material grade (e.g., in compliance with IEC 60079-0 Appendix A), certification information (e.g., existing UL, ATEX, CCC certificates, etc.) or technical parameters (such as CTI value, flame retardancy, chemical resistance, glass transition temperature Tg, etc.);

PCB BOM – Component Summary

Component	Part / Value	Qty	Manufacturer	MPN
Capacitor	1 µF	1	Samsung	CL21B105KBFNNE
Capacitor	100 nF	2	Yageo	CC0805KRX7R9BB104
Capacitor	10 µF	4	Murata	GRM21BR61H106KE43L
Capacitor	33 µF	2	TDK	C2012X5R1A336MTJ00E
Capacitor	47 nF	1	Samsung	CL10B473KB8NNNC
Capacitor	22 nF	1	Samsung	CL10B223KB8NNNC
Capacitor	33 pF	1	Samsung	CL10C330JB8NNNC
Capacitor	100 nF	2	Yageo	CC0603KRX7R9BB104
Capacitor	100 nF	38	Samsung	CL05B104KO5NNNC
Connector	XH-2A, Black	2	Boomele	XH-2A
TVS Diode	SMBJ33A	1	Ruilon	SMBJ33A
Zener Diode	LBZT52C12T1G	1	LRC	LBZT52C12T1G
Schottky Diode	BAT54C	2	CJ	BAT54C
ESD Protection Diode	LES5D5.0CT1G	3	UMW	LES5D5.0CT1G
TVS Diode	SM712	1	DOWO	SM712
Schottky Diode	SS54	2	MDD	SS54
Schottky Diode	SK36	1	Good-Ark	SK36
Fuse	MINISMDC260F/16	2	Littelfuse	MINISMDC260F/16-2
RF Connector	U.FL	1	—	—
Input Connector	VIN	1	—	—
Ferrite Bead	300 Ω	1	TDK	ACM7060-301-2PL-TL01
Inductor	2.2 µH	1	SXN	SMDRH105R-2R2NT
Inductor	2.2 µH	1	APV	ANR3015T2R2M
Ferrite Bead	100 Ω	1	TDK	MPZ2012S101AT000
Inductor	10 µH	1	SXN	SMDRH105R-100MT
LED	19-217/BHC-ZL1M2RY/3T	1	Everlight	19-217/BHC-ZL1M2RY/3T
Transistor	UMH3NTN	1	ROHM	UMH3NTN
MOSFET	AO4407	1	JSMSEMI	AO4407
MOSFET	SI7465DP-T1-GE3	1	Vishay	SI7465DP-T1-GE3
Resistor	187 kΩ	1	UNI-ROYAL	0603WAF1873T5E
Resistor	82 kΩ	1	UNI-ROYAL	0603WAF8202T5E
Resistor	19.1 kΩ	1	UNI-ROYAL	0603WAF1912T5E
Resistor	1 kΩ	1	UNI-ROYAL	0603WAF1001T5E
Resistor	56 kΩ	2	UNI-ROYAL	0603WAF5602T5E
Resistor	189 kΩ	1	Yageo	RT0603BRD07189KL
Resistor	33 kΩ	1	UNI-ROYAL	0603WAF3302T5E
Resistor	470 Ω	1	UNI-ROYAL	0603WAF4700T5E
Push Button Switch	SKQYAAE010	1	Alps Alpine	SKQYAAE010
Boost Converter IC	TPS61175PWP	1	Texas Instruments	TPS61175PWP
Buck Converter IC	LMR51450SDRRR	1	Texas Instruments	LMR51450SDRRR
Inductor	4.7 µH	1	ABC	GSSM06304R7M2AU

Ex Safety-Critical Component BOM

Ref	Ex-relevant function	Components (MPN, Qty)	Key parameters to document	Status
EL-01	Overcurrent protection	Littelfuse MINISMD260F/16-2 (PTC), 2	I-hold 2.6 A, I-trip 5.0 A, V-max 16 V, trip time, temperature rating	Present.
EL-02	Input transient/overvoltage protection	Ruilon SMBJ33A, 1	V-RWM 33 V, V-clamp, P-pk 600 W	Present
EL-02a	Communication interface protection	DOWO SM712, 1 (RS-485); UMW LESD5D5.0CT1G, 3 (ESD)	Clamp voltage, protected lines	Present, medium priority
EL-03	Reverse-polarity protection	Vishay SI7465DP-T1-GE3 (P-MOSFET), 1 + LRC LBZT52C12T1G (gate clamp zener), 1	V-DS, I-D, R-DS(on), P-diss, V-GS max	Confirm from schematic
EL-03a	OR-ing / blocking diodes	MDD SS54, 2; Good-Ark SK36, 1	V-RRM, I-F, V-F, dissipation	Confirm function (may be the boost diode)
EL-04	Buck converter, possible hot surface	TI LMR51450SDRRR, 1 + feedback divider	V-in max, I-out, switching frequency, max temperature at full load and under fault	Present
EL-05	Boost converter, possible hot surface	TI TPS61175PWP, 1 + feedback divider	V-in, V-switch max, I-switch limit, V-out setpoint	Present
EL-06	Power and filter inductors, possible hot surface	SXN SMDRH105R-2R2NT; APV ANR3015T2R2M; SXN SMDRH105R-100MT; ABC GSSM06304R7M2AU; TDK ACM7060-301-2PL-TL01 (input CM choke); TDK MPZ2012S101AT000	I-sat, I-rms, DCR, temperature rise (ΔT) at max current	Present. Map each inductor to its converter
EL-07	MQ heater supply	Candidate: JSMSEMI AO4407 (P-MOSFET, 30 V)	Heater V/I, switching method, current limiting	Unconfirmed.
EL-08	Alarm driver (buzzer/LED)	ROHM UMH3NTN, 1; Everlight 19-217/BHC-ZL1M2RY/3T, 1; 470 Ω resistor	Load current	Buzzer not in BOM
EL-09	Terminals / connectors	Boomele XH-2A, 2; VIN connector, 1	Current/voltage rating, retention type (anti-loosening)	VIN connector MPN blank
EL-10	Switching contact	Alps Alpine SKQYAAE010, 1	Contact rating	Present

c. Material specification sheets / data sheets

Requirement: Material specification sheets/data sheets: Detailed technical parameters for non-metallic materials (enclosures, seals, insulators, potting compounds, etc.) to prove compliance with standard requirements (such as heat and cold resistance, anti-aging, anti-static, flame retardancy, CTI value, comparative tracking index, chemical resistance, etc.). Provide test reports or conformity declarations from material suppliers;

Non-metallic materials used in the GLD that may affect explosion-protection performance, electrical insulation, environmental sealing, electrostatic behavior, or long-term durability are identified and controlled by material grade and supplier specification.

Relevant materials include enclosure gaskets and seals, cable-entry sealing components, PCB laminate, connector housings, wire insulation, antenna insulating parts, fan polymeric parts, and any adhesive, sealant or potting material used in the final production configuration.

For each applicable material, the manufacturer's technical datasheet or conformity declaration shall provide the relevant properties, including service-temperature range, heat and cold resistance, ageing resistance, flame retardancy, CTI where applicable, electrostatic properties where applicable, chemical resistance, UV/weather resistance, and glass-transition temperature where relevant.

Materials for which supporting documentation is not yet available shall remain identified as open items until the required supplier documentation or test evidence has been obtained.

d. Manufacturing process description

Requirement: Manufacturing process description: Especially for processes that affect explosion-proof safety (such as enclosure processing accuracy control, explosion-proof surface treatment, welding process, potting process, die-casting process, bonding process, etc.);

e. Explosion-proof related calculations and explanations (if applicable)

e.1 Gas access path and internal arrangement

Gas access path to the MQ sensors

Ambient gas enters the detector through the stainless-steel mesh located at the front of the enclosure. During the sampling cycle, the internal DC fan draws ambient gas through this front mesh into the sensing chamber. Eight MQ-series gas sensors are installed inside the sensing chamber. Each MQ sensor incorporates its own stainless-steel protective mesh surrounding the sensing element. Therefore, the sampled gas passes through two successive mesh structures before reaching the sensing element: first the front enclosure mesh, and subsequently the integrated stainless-steel mesh of the individual MQ sensor.

e.2 Identification and assessment of potential ignition sources

Ref.	Potential ignition source	Possible ignition mechanism	Exposure to explosive atmosphere	Existing protection / design feature	Additional assessment required	Status
IS-01	MQ sensor internal heater	Hot surface due to internal heating element	Sampled gas reaches the MQ sensing element after passing through the front enclosure mesh and the integrated stainless-steel protective mesh of each MQ sensor	Two successive mesh structures exist between ambient atmosphere and the sensing element: front enclosure mesh + individual MQ sensor mesh	Verify maximum sensor/heater temperature under normal and fault conditions. Obtain manufacturer documentation confirming the construction and, if applicable, any defined flame-arresting function of the MQ sensor mesh	Open

Ref.	Potential ignition source	Possible ignition mechanism	Exposure to explosive atmosphere	Existing protection / design feature	Additional assessment required	Status
IS-02	DC sampling fan	Motor winding heating, electronic commutation, internal electrical fault, stalled-rotor heating	Directly located in the gas sampling path, immediately behind the front stainless-steel mesh	Front stainless-steel mesh is located upstream of the fan	Confirm fan type and construction; verify rated voltage/current, maximum surface temperature, stalled-fan behavior, fault condition, and applicable Ex protection or component certification	Open
IS-03	Main motherboard PCB	Electrical arc/spark from energized circuitry, component failure, short circuit, localized overheating	To be confirmed. Current construction shows the motherboard and MQ sensors installed inside the same enclosure volume	Enclosure provides mechanical protection; final Ex protection concept still under evaluation	Determine whether sampled gas can reach the motherboard volume. Assess energized circuits, maximum voltages/currents, fault energy, and protection concept	Open
IS-04	DC/DC power-conversion circuits	Semiconductor overheating, inductor heating, switching fault, short circuit	Depends on whether the PCB volume is gas-accessible	Input includes PTC, TVS and MOSFET-based protection according to the current design	Thermal assessment under maximum load and fault conditions; verify protection coordination and component ratings	Open
IS-05	24 VDC input terminals and internal connectors	Make/break spark, loose-contact heating, short circuit	Depends on gas accessibility to the terminal compartment / internal enclosure	Mechanical enclosure and cable-entry system	Confirm terminal location relative to gas-accessible volume; verify terminal rating, conductor retention, torque requirements, creepage/clearance and approved entry arrangement	Open
IS-06	LED / audible alarm module	Internal electrical switching, component heating, possible electrical fault	Depends on alarm-module construction and whether its internal electrical parts communicate with hazardous atmosphere	Separate alarm-module housing shown in the mechanical assembly	Obtain module construction and certification data; confirm whether Ex-certified or included within the equipment protection concept	Open
IS-07	LoRa module / ESP32-S3 / digital electronics	Semiconductor overheating or electrical fault	Depends on gas accessibility to the main PCB volume	Installed within the main enclosure	Assess maximum component temperature and credible electrical fault conditions if gas can access this volume	Open
IS-08	RF transmitter / antenna connection	RF electromagnetic energy and connector-related electrical effects	External antenna and internal RF circuitry form part of the complete system	Firmware transmit power is limited to the configured operating range	Document maximum RF output power, antenna gain and approved antenna configuration; assess against applicable Ex requirements	Open
IS-09	PCB inductors and magnetic components	Surface heating under overload or converter fault	Depends on gas accessibility to PCB volume	Normal current-limited converter operation	Measure or calculate worst-case temperature at maximum load, low/high input voltage and relevant converter fault conditions	Open
IS-10	MOSFETs, power diodes and protection devices	Localized overheating during overload, surge or reverse-polarity event	Depends on gas accessibility to PCB volume	PTC fuse, TVS and MOSFET input-stage protection are included in the design	Verify component ratings and maximum temperature during abnormal conditions; confirm protective-device coordination	Open
IS-11	Electrostatic charging of non-metallic parts	Electrostatic discharge	External non-metallic surfaces such as gasket, antenna insulation, cable-gland parts and other exposed polymeric components	Metallic enclosure includes an external grounding stud	Identify all exposed non-metallic materials and dimensions; assess electrostatic charging risk and material properties	Open

Ref.	Potential ignition source	Possible ignition mechanism	Exposure to explosive atmosphere	Existing protection / design feature	Additional assessment required	Status
IS-12	Metallic enclosure and mechanical joints	Spark due to poor bonding, impact, or potential difference	External hazardous atmosphere	Metal enclosure with external protective-earth connection	Verify electrical bonding continuity between enclosure parts, grounding connection, corrosion control, and fastening method	Open
IS-13	Cable entry / gland / blanking devices	Loss of protection due to incorrect gland, loose fitting, or unsuitable entry component	Direct interface between external hazardous atmosphere and enclosure	Dedicated cable-entry arrangement provided	Finalize approved gland/adaptor/blanking-device list; verify thread type, engagement, torque, IP and Ex certification	Open
IS-14	Front stainless-steel mesh assembly	Potential flame transmission path if an internal ignition occurs	Direct interface with ambient hazardous atmosphere	Stainless-steel front mesh installed as first barrier at the gas inlet	Confirm manufacturer, part number, construction, installation method and certification. If claimed as flame-arresting, provide certificate/test evidence supporting that function	Open
IS-15	MQ sensor stainless-steel protective mesh	Potential flame-transmission path between internal MQ sensing/heater region and sensing chamber	Directly surrounding each MQ sensing element	Integrated protective stainless-steel mesh is provided by each MQ sensor	Obtain manufacturer construction data. Do not claim ATEX flame-arresting function unless explicitly supported by manufacturer documentation or certification	Open
IS-16	Mechanical impact/friction	Hot particle or mechanical spark from damaged/moving metallic parts	External and internal depending on location	Fixed installation, fan is the principal moving component	Confirm fan clearance, mounting retention and resistance to mechanical contact under foreseeable conditions	Open
IS-17	Battery circuit / battery compartment, if retained in certified product	High fault current, cell heating, connector spark	Depends on final certified product configuration	Current draft states continuous 24 VDC production configuration	Confirm whether battery hardware is excluded, depopulated, disabled or included in certification scope. If excluded, remove it from certified-product drawings/BOM as appropriate	Open

Preliminary assessment:

Preliminary ignition-source assessment

The GLD contains several potential electrical and thermal ignition sources, including the MQ-series sensor heaters, DC sampling fan, energized PCB circuitry, power-conversion components, terminals, alarm electronics, RF circuitry, and electrostatic discharge from applicable non-metallic parts.

Ambient gas enters the detector through the front stainless-steel mesh and is drawn into the sensing chamber by the DC fan during the sampling cycle. Each MQ sensor incorporates an additional stainless-steel protective mesh surrounding its internal sensing element. The MQ sensing element is therefore located behind two successive mesh structures: the front enclosure mesh and the individual MQ sensor mesh.

The final ignition-risk assessment depends on confirmation of the internal gas-accessible volume, maximum surface temperatures, credible fault conditions, component certification, and the final explosion-protection concept selected for the equipment.

e.3 Explosion-protection concept and protective construction

e.3.1 Overall explosion-protection concept

The GLD is intended for installation in hazardous gas areas and is designed so that potential ignition sources are either isolated from the external explosive atmosphere, enclosed within an appropriate protective construction, or otherwise controlled so that they cannot ignite the surrounding atmosphere.

Ambient gas is intentionally admitted into the sensing section of the detector for measurement purposes. Therefore, the explosion-protection design distinguishes between gas-accessible sensing areas and protected electrical/electronic areas.

e.3.2 Gas-accessible volume and protected volume

The front sensing chamber is intentionally gas-accessible. Ambient gas enters through the front stainless-steel mesh and is

drawn inward by the DC sampling fan. The eight MQ sensors are located in this gas-accessible sensing region.

The main electronic PCB is separated from the gas-accessible sensing chamber by pcb plate directly, preventing sampled gas from reaching the main energized electronics.

e.3.3 Front stainless-steel mesh protection

The front stainless-steel mesh forms the first protective element in the gas inlet path. It allows ambient gas to enter the sensing chamber while providing mechanical protection and, where supported by component certification, a flame-arresting function.

e.3.4 MQ sensor protective construction

Each MQ sensor includes an integrated stainless-steel protective mesh surrounding the internal sensing and heating structure. Therefore, gas entering the sensing chamber must pass through the individual MQ sensor mesh before reaching the sensing element.

The protective function of this integrated mesh shall be based on the MQ manufacturer's technical documentation. No flame-arresting or Ex protection claim shall be assigned to the MQ sensor mesh unless supported by appropriate documentation or certification.

e.3.5 Protection of the DC fan

The DC sampling fan is located behind the front stainless-steel mesh and therefore operates within the gas sampling path. Its ignition risk is addressed through the selected fan construction and the overall explosion-protection concept. Fan surface temperature, winding/electronic temperature, stalled-rotor behavior, and internal commutation shall be assessed.

e.3.6 Protection of main electronics

Energized electronic circuits, including the main PCB, power-conversion circuits, processing electronics, communication electronics, and internal connectors, are treated as potential ignition sources. Their protection depends on the final confirmed protection concept and on whether these circuits are located within a gas-accessible volume.

e.3.7 Flame transmission paths

All potential flame-transmission paths between the internal enclosure volume and the external hazardous atmosphere shall be identified and controlled by approved construction features. These include the front gas inlet assembly, enclosure cover joint, threaded interfaces, cable entries, antenna interface, alarm-module connection, and any other penetration through the enclosure wall.

e.3.8 Cable entries, antenna, alarm module

All enclosure penetrations shall use components compatible with the final Ex protection concept, gas group, temperature range, IP rating, and mechanical thread configuration.

e.3.9 Grounding and electrostatic protection

The metallic enclosure is bonded to protective earth through the designated external grounding stud. Conductive enclosure parts shall maintain electrical continuity to minimize electrostatic charging and potential differences.

e.3.10 Temperature protection

Surface temperatures of the MQ sensors, fan, power-conversion components, enclosure, alarm module, and other relevant parts shall be evaluated under maximum operating load and applicable fault conditions. The final temperature class shall be assigned only after temperature calculations and/or testing have been completed.

e.3.11 Fault and abnormal conditions

The protective concept shall remain effective under normal operation and under applicable foreseeable abnormal or fault conditions, including fan stall, blocked airflow, heater abnormality, power-converter failure, short circuit, loose terminal connection, and maximum permitted ambient temperature.

f. Temperature group calculation (estimation of the hottest point temperature)

6.f.1 Proposed temperature class

The current proposed temperature class for the GLD is T4. Final assignment of the temperature class shall be based on the measured and/or calculated maximum temperature of all relevant equipment surfaces and components under the applicable operating and fault conditions.

6.f.2 Maximum ambient temperature

Maximum ambient temperature for assessment: +85 °C

6.f.3 Candidate hottest points

Ref.	Component / Location	Reason for Assessment
T-01	MQ sensor heater / sensor head	Heater is continuously energized during sensor operation
T-02	DC fan motor and drive electronics	Heat generation during sampling operation and under stalled-rotor conditions
T-03	5 V regulator / power converter	Supplies the MQ sensor heater
T-04	3.3 V regulator	Supplies the MCU and digital electronics
T-05	Boost/buck converter ICs	Switching losses
T-06	Power inductors	Copper and core losses
T-07	Input MOSFET / protection devices	Conduction losses and heating under fault conditions
T-08	ESP32-S3	Processing load, including on-device AI inference
T-09	LoRa module	Heat generation, particularly during RF transmission
T-10	LED/buzzer alarm module	Heat generation while the alarm is active
T-11	Main enclosure external surface	Determines the external surface temperature for temperature class (T-class) assignment
T-12	Front mesh / cover near sensor chamber	Influenced by MQ heater output and airflow conditions

g. Usage and installation instructions

The GLD shall be installed, commissioned, operated, inspected, and maintained only in accordance with the approved product configuration and installation instructions. The instructions define the permitted installation environment, mechanical mounting, electrical supply, cable entry, grounding, antenna arrangement, commissioning procedure, operating modes, alarm handling, cleaning, periodic inspection, maintenance, repair restrictions, and control of firmware/configuration changes.

a) Clear safety warnings

- i. Only trained and authorized personnel may install, commission, operate, inspect, or maintain the detector.
- ii. Isolate and verify the 24 VDC supply before opening the enclosure or changing wiring. Apply the site lockout/tagout procedure.
- iii. Use only the approved production power arrangement: 24 VDC.
- iv. Do not open the enclosure while an explosive atmosphere may be present unless the equipment is de-energized and the area is confirmed gas-free.
- v. Perform alarm tests with appropriate site controls and hearing protection where audible

devices are connected.

- vi. Perform sensor nulling only in confirmed clean air. Accept the operation only when all eight sensors report a complete 8/8 result; otherwise correct the cause and repeat.
- vii. Do not drill, weld, enlarge cable entries, bypass protective devices, or modify the enclosure without written engineering and Ex approval — any unauthorised modification to an approved enclosure invalidates the certificate.
- viii. Remove the unit from service if there is visible damage, water ingress, an unresolved sensor fault, failed alarm proof, or a loss of protective-earth continuity.

b) Permitted installation environment

The Gas Leak Detector (GLD) is intended for fixed installation in hazardous areas associated with oil and gas refinery facilities, including process units, tank farms, pipe racks, and storage/loading-unloading areas in which flammable gas atmospheres may be present.

The current certification target is **Equipment Group II, Category 2G / EPL Gb**, for installation in **Zone 1** gas hazardous areas, subject to confirmation by the ExCB. The proposed gas group is **IIC**, and the proposed temperature class is **T4**. Final confirmation of the gas group and temperature class shall be based on the completed explosion-protection assessment, applicable component documentation, and the results of the required temperature evaluation and testing.

The GLD is intended to operate within an ambient temperature range of **–20 °C to +85 °C** and an operating relative-humidity range of **5–95% RH, non-condensing**, as defined in the current product specification.

Installation shall be limited to locations and environmental conditions that are within the certified equipment marking and the conditions of use stated in the final certificate and approved instruction manual. The equipment shall not be installed in a hazardous area having a more severe gas group, temperature classification, ambient-temperature range, or other environmental condition than those covered by the final certification.

The GLD is intended for **gas hazardous-area service only** within the current certification scope. Installation in combustible-dust hazardous areas is excluded unless such use is separately assessed, tested, and approved as part of the certification scope.

The detector shall be installed in a position that permits free access of the surrounding atmosphere to the front gas-sampling inlet. The stainless-steel gas-entry mesh and the associated sampling path shall not be obstructed, covered, sealed, painted, or otherwise modified in a manner that could impair gas access to the sensing section.

The unit shall be mounted as a fixed installation using the approved mounting arrangement and in accordance with the orientation and mechanical details specified in the controlled drawings. Adequate clearance shall be maintained around the gas-sampling inlet, cable entries, antenna interface, and enclosure joints to permit correct operation, inspection, and maintenance.

The installation location shall protect the equipment from mechanical damage, unauthorized modification, excessive contamination, and environmental conditions beyond the approved operating limits. Cable entries, grounding connections, antenna interfaces, and other enclosure penetrations shall be installed only in accordance with the approved product configuration and the applicable installation instructions.

c) Mechanical mounting

The Gas Leak Detector (GLD) shall be installed as a fixed-mounted device using the approved mechanical mounting bracket provided for the product. The mounting arrangement shall correspond to the configuration defined in the controlled mechanical drawings and shall not be modified without engineering review and approval.

The GLD is intended to be attached to existing plant structures by means of the approved bracket arrangement. Installation shall not require drilling, welding, machining, or other permanent modification of the supporting plant structure unless such modification is separately approved by the site owner and is outside the certified GLD construction.

The mounting bracket, fasteners, clamps, and associated hardware shall be of the specified type, material, size, and mechanical strength defined in the approved production drawings and bill of materials. Substitution of mounting components that may affect the mechanical integrity or certified configuration of the equipment shall not be permitted without controlled engineering approval.

The detector shall be mounted securely so that it cannot become loose, rotate unintentionally, or move under normal service conditions, vibration, maintenance activity, or foreseeable external mechanical loading. All mounting fasteners shall be tightened in accordance with the torque values specified in the approved assembly or installation documentation.

The installed orientation shall conform to the orientation specified in the controlled mechanical drawings. The installation shall maintain unobstructed access to the gas-sampling inlet, front stainless-steel mesh, antenna interface, cable-entry points, grounding connection, and enclosure joints required for normal operation, inspection, and maintenance.

The gas-sampling inlet shall not be positioned directly against a wall, pipe, mounting plate, structural member, or other obstruction that could restrict the flow of ambient gas into the sensing area. Adequate clearance around the sampling inlet shall be maintained in accordance with the approved installation drawing.

The mounting location shall be selected so that the enclosure and its associated components are protected from foreseeable mechanical impact, excessive vibration, deformation, and other mechanical stresses that could impair the integrity of the enclosure, gas-entry assembly, cable entry, antenna interface, alarm module, or grounding connection.

After installation, the mounting assembly shall be visually inspected to confirm correct orientation, secure fastening, absence of enclosure deformation, and unobstructed access to the sensing inlet and service interfaces.

d) Gas inlet / sensing-area requirements

The Gas Leak Detector (GLD) shall be installed and maintained so that the gas-sampling inlet and sensing area remain continuously accessible to the surrounding atmosphere during normal operation.

Ambient gas enters the detector through the stainless-steel mesh located at the front of the enclosure. During the sampling cycle, the internal DC fan assists the movement of ambient gas through the front mesh and into the sensing area. Eight MQ-series gas sensors are installed within this sensing area. Each MQ sensor incorporates an additional stainless-steel protective mesh surrounding its sensing element.

The front gas-entry mesh, sampling opening, and associated airflow path shall remain unobstructed. The gas inlet shall not be covered, sealed, painted, taped, coated, or otherwise modified in a manner that could restrict or alter the intended gas flow to the sensing area. No additional filter, membrane, screen, protective cover, or other material shall be installed over the gas inlet unless specifically approved as part of the certified product configuration.

The detector shall be installed with sufficient clearance around the gas-sampling inlet to allow free circulation of the surrounding atmosphere. The front sensing inlet shall not be positioned directly against a wall, pipe, mounting plate, structural member, insulation, or other object that could restrict gas entry or interfere with the intended sampling airflow. The required minimum clearance, where applicable, shall be defined in the controlled installation drawing.

The stainless-steel front mesh shall be maintained in its approved production configuration. It shall not be removed, replaced, drilled, enlarged, reshaped, or substituted with a different mesh or protective element unless the change has been evaluated and approved through the applicable engineering and Ex certification change-control process.

The integrated protective mesh of each MQ sensor shall also remain intact and free from mechanical damage, deformation, contamination, or blockage. MQ sensors with damaged or missing protective mesh shall not be used.

The sensing area shall be kept free from excessive dust, deposits, oil, paint, corrosion products, insects, or other contamination that could obstruct the front mesh, fan, individual sensor meshes, or the intended gas-sampling path. Cleaning shall be performed only in accordance with the approved maintenance instructions and shall not damage the mesh, fan, sensor surfaces, enclosure sealing features, or other Ex-critical components.

The DC sampling fan shall be installed in the approved position and orientation behind the front stainless-steel mesh. The airflow direction, fan mounting, electrical connection, and mechanical clearance shall conform to the controlled assembly drawing. The fan shall not be substituted with a different model without controlled engineering review and approval.

The gas-sampling path shall be considered part of the controlled product construction. Any modification that changes the

geometry, mesh arrangement, fan position, airflow path, sensing-chamber volume, or position of the MQ sensors shall require evaluation before implementation.

During inspection and maintenance, the front mesh, fan, sensing area, and individual MQ sensor meshes shall be checked for blockage, damage, contamination, corrosion, or other conditions that could affect gas access or the approved protection concept. If any such condition is detected, the detector shall be removed from normal service until the condition has been corrected and the unit has been verified as suitable for operation.

Note: The minimum clearance in front of the gas inlet, the approved fan orientation, and any dimensional requirements of the sensing chamber shall be specified in the controlled mechanical and installation drawings.

e) Detailed installation requirements

Installation shall comply with the approved equipment documentation, applicable Ex requirements, and site installation procedures.

Cable Entry Method

Only cable glands, adaptors, blanking elements, and sealing components approved for the specified protection concept, certification, thread type, IP rating, cable outer diameter, and installation temperature range shall be used.

The following requirements shall apply:

- Do not create additional cable entries or enlarge existing entries.
- All unused cable entries shall be fitted with approved Ex-certified blanking elements.
- General-purpose cable glands shall not be substituted for approved Ex components.
- Follow the manufacturer's specified sealing, assembly, and clamping sequence.
- Record the installed component part number and applicable certificate reference.
- Threaded covers and entries shall be fully engaged to the minimum number of threads specified by the applicable enclosure certification before the locking mechanism is secured.
- Cable glands and entries shall maintain the required enclosure IP rating.

Cable Gland Selection Guide

Hazardous Area	Required Gas Group	Recommended Cable Gland Type
Zone 1 / Zone 21	IIC	Ex d barrier gland required
Zone 2 / Zone 22	IIA / IIB	Any suitable Ex d gland permitted

This selection should remain subject to the final enclosure certificate and applicable installation requirements. The reference documentation likewise specifies Ex d barrier glands for hazardous areas requiring IIC apparatus and permits suitable Ex d glands for IIA/IIB applications.

Torque Requirements

The enclosure fasteners shall be tightened using a 22 mm wrench/socket. Final tightening shall be performed in accordance with the approved mechanical assembly procedure and manufacturer-specified torque requirements.

Where applicable, the installation record shall document the verified tightening condition.

Grounding Requirements

The enclosure is provided with a dedicated external grounding terminal located on the side of the case.

- Connect the external grounding terminal to the site's approved Protective Earth (PE) system.
- Use an appropriate grounding conductor and approved cable lug suitable for the grounding terminal.

- Ensure the grounding connection is mechanically secure and protected against loosening and corrosion.
- Verify grounding continuity before energizing the equipment.
- The external grounding connection shall remain accessible for inspection and maintenance.

Cleaning Requirements

- For routine cleaning, isolate the equipment as required by the applicable site procedure and use a soft, lint-free cloth lightly dampened with clean water or an approved mild cleaning agent.
- Do not use abrasive materials, aggressive solvents, dry compressed air, high-pressure water or spray, or tools that may damage the enclosure, gasket, nameplate, antenna, or cable entries.
- Prevent liquids from entering the enclosure through any opening, joint, or cable entry.

f) Operating instructions and maintenance requirements

Requirement: *Operating instructions, maintenance requirements (frequency, content, precautions);*

Operating Modes

Mode	Purpose	Use Restriction
Inference	Normal gas-class inference and alarm operation	Production operating mode after commissioning
Dataset Collection	Engineering and data-collection activities	Not a substitute for approved detection operation
Nulling	Sensor baseline / calibration activity	Clean air only; perform under the approved procedure and require a complete 8/8 result

Operating Instructions

- Before each operating period, verify the power supply, enclosure condition, diagnostic status, communication status, and absence of active faults.
- An alarm shall be treated as a potential process-safety event until the applicable site response procedure determines otherwise.
- If the detector reports a fault or loses required communication, the responsible operator shall be notified and the affected detection point shall be placed under the site's impairment management procedure.
- Firmware, model files, alarm thresholds, calibration parameters, or other safety-related configuration shall not be modified in the field without controlled authorization, documented change management, and an appropriate rollback procedure.

Periodic Inspection & Maintenance Schedule

Interval	Activities
Daily / Each Shift	Check power supply, enclosure condition, diagnostic status, LoRa communication status, and ensure there are no active faults.
Every 3 Months	Perform visual inspection of the sensor and protective mesh. Applicable particularly to electrochemical/catalytic gas sensors, including MQ-type sensors that use chemical sensing elements and heaters.

Interval	Activities
Every 6 Months	Perform alarm function testing (visual/audible), sensor recalibration/nulling, and visual inspection of flame paths and cover threads for signs of corrosion or damage.
Annually (Minimum)	Conduct a comprehensive inspection, including grounding continuity verification, cable gland and terminal torque checks, gasket/seal condition assessment, and evaluation of sensor element service life for replacement planning.

h. Nameplate information

Requirement: Nameplate information (must include all ATEX required marking information) ;

Proposed ATEX Nameplate Marking

The equipment nameplate shall include all applicable ATEX and hazardous-area identification information, including:

- ATEX Equipment Category: II 2 G D
- Gas Protection: Ex db IIC T4 Gb
- Dust Protection: Ex tb IIIC T135°C Db
- Ambient Temperature: Tamb = –40°C to +85°C
- CE Marking and Notified Body Number
- Equipment identification / model
- Manufacturer name and/or identification
- Certificate number

The proposed protection marking is based on the reference flame detector documentation, which specifies II 2 G D, Ex db IIC T4 Gb, Ex tb IIIC T135°C Db, Tamb –40°C to +85°C.

i. Ex components ATEX certificates

Proposed ATEX Certification

The proposed equipment certification shall meet the following hazardous-area protection classification:

II 2 G D

Ex db IIC T4 Gb

Ex tb IIIC T135°C Db

Tamb = –40°C to +85°C

The selected Ex components, including the flameproof enclosure, cable glands, and other applicable Ex-certified components, shall have valid certificates appropriate to the intended hazardous-area classification.

For reference, the source document identifies the flame detector enclosure as ATEX Category II 2 G D, with Ex db IIC T4 Gb and Ex tb IIIC T135°C Db, and references ATEX certification under Directive 2014/34/EU.

3. Sample Information

1) Sample model, serial number, and status

Requirement: Provide the model, serial number, and status (such as whether it can be powered on and run) of the sample;

Model	GLD_V2
Serial Number	GLD2-0x1001
Status	Inference / Normal Operation

2) Test fixtures and auxiliary equipment

Requirement: Provide necessary test fixtures or auxiliary equipment.

Equipment	Purpose
Regulated 24 VDC power supply	Provides a stable 24 VDC supply to the GLD during commissioning and functional testing.
Digital multimeter	Measures the actual voltage, verifies polarity and continuity, and supports grounding/bonding verification.
Calibrated gas source / gas test kit	Provides the required gas types to verify that the sensors respond correctly when exposed to the target gases.
Gas test chamber	Provides a controlled gas-exposure environment so that gas concentration, exposure time, and test conditions can be reproduced.
Alarm load / relay simulator	Physically verifies that the alarm output and connected electrical load operate correctly.
LoRa peer / gateway	Verifies that data transmitted by the GLD is successfully received by the intended system.
Laptop and approved software	Used to review diagnostic information, configuration, logging, firmware identification, and to store commissioning and test evidence.
Calibrated torque screwdriver	Ensures that glands, terminals, enclosure fasteners, and grounding connections are tightened to the approved torque values.

Appendix – IECEx/ATEX 认证信息提供要求

1. 基本资料 (申请与组织):

- 1) 申请表 (由ExCB提供模板)。
- 2) 制造商营业执照/公司注册证明。
- 3) 制造商组织结构图、联系信息。
- 4) 制造工厂地址、生产设施简介。
- 5) (如适用) ISO 9001 证书及质量手册、程序文件目录 (用于QAR评审)。

2. 技术文件 (核心):

- 1) 产品详细描述：
- 2) 产品名称、型号、规格清单。
- 3) 完整、清晰的功能描述和技术参数 (电气参数、机械参数等)。
- 4) 清晰的产品照片 (整体和关键部件)。
- 5) 预期用途和安装环境描述 (如适用气体组别IIC/IIB/IIA, 温度组别T1-T6, 环境温度范围, 区域划分0/1/2或20/21/22)。
- 6) 设计与制造信息：
 - a. 完整的图纸：总装图、部件图、电路图、电气原理图、PCB布局图 (如适用)、外壳结构图 (需标明关键隔爆参数如间隙、长度、容积)、接线盒图、端子图、接地连接图等。图纸需清晰标注尺寸、公差、材料。
 - b. 关键部件清单 (BOM)：列出所有影响防爆安全的零部件 (如外壳、衬垫、端子、电缆引入装置、开关、光源、电池、灌封胶、塑料件、印制电路板等)，包括制造商、型号规格、材料等级 (如符合IEC 60079-0附录A)、认证信息 (如已有UL, ATEX, CCC等证书) 或技术参数 (如CTI值、阻燃等级、耐化学性、玻璃化温度T_g等)。
 - c. 材料规格书/数据表：特别是非金属材料 (外壳、密封件、绝缘件、灌封胶等) 的详细技术参数，证明其符合标准要求 (如耐热耐寒、抗老化、抗静电、阻燃性、CTI值、相比漏电起痕指数、耐化学腐蚀性等)。提供材料供应商的测试报告或符合性声明。
 - d. 制造工艺描述：特别是对防爆安全有影响的工艺 (如外壳加工精度控制、隔爆面处理、焊接工艺、浇封工艺、压铸工艺、粘接工艺等)。
 - e. 防爆相关计算与说明 (如适用)：

- f. 温度组别计算（最热点温度估算）。
 - g. 使用与安装说明(草案)：
 - a) 清晰的安全警告。
 - b) 详细的安装要求（如电缆引入方式、扭矩要求、接地要求、清洁要求）。
 - c) 操作说明,维护要求（周期、内容、注意事项）。
 - h. 铭牌信息（需包含所有IECEx要求的标志信息）。
 - i. 防爆元器件ATEX证书
- 7) 样品信息：
- a. 提供样品的型号、序列号、状态（如是否可通电运行）。
 - b. 提供必要的测试工装或辅助设备。